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Pharmacology of Eicosanoids

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INTRODUCTION

Autacoids are substances that are rapidly synthesized in response to specific stimuli, act quickly in the immediate environment, and remain active for only a short time before degradation. **Eicosanoids** represent a chemically diverse family of autacoids that are mostly derived from arachidonic acid. Research on eicosanoids continues to elucidate their critical roles in inflammatory, neoplastic, and cardiovascular physiology and pathophysiology. Numerous pharmacologic interventions in eicosanoid pathways—including the nonsteroidal anti-inflammatory drugs (NSAIDs), cyclooxygenase-2 (COX-2) inhibitors, leukotriene inhibitors, and others—are useful in the clinical management of inflammation, pain, and fever. Given the diverse bioactivities of eicosanoids, future research in eicosanoid physiology and pharmacology may lead to the development of new therapeutics for the treatment of asthma, inflammatory diseases, autoimmune diseases, glomerulonephritis, cancer, cardiovascular diseases, and other clinical conditions.

PHYSIOLOGY OF EICOSANOID METABOLISM

Eicosanoids are centrally involved in a number of metabolic pathways that exhibit diverse roles in inflammation and cellular signaling. The vast majority of these pathways center on reactions involving the metabolism of arachidonic acid (Fig. 42-1). The following section considers the biochemical steps leading to arachidonic acid generation, and then discusses the cyclooxygenase, lipoxygenase, epoxygenase, and isoprostane pathways of arachidonic acid metabolism. The word *eicosanoid* arises from the Greek root for twenty, and the term classically refers to 20-carbon molecules derived from arachidonic acid oxygenation. The term eicosanoid also applies broadly to various other molecules—such as resolvins, protectins, and maresins—that are derived from docosahexaenoic acid, a 22-carbon precursor. The term *docosanoids* is sometimes also used to describe these 22-carbon structures.

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CASE

Ms. G, a 49-year-old Native American female, visits her physician because of joint pain and chronic fatigue. Her history reveals general joint stiffness and pain for the past 3 weeks, especially in the early morning and worst in the metacarpophalangeal and proximal interphalangeal joints. Ms. G is advised to take ibuprofen as needed, and this medication provides relief of her pain for some time.

Six months later, Ms. G notes indigestion and new vomiting of "coffee-grounds"-like material. Her physician recommends an upper gastrointestinal endoscopic examination, which reveals gastric mucosal erosion and hemorrhage. Based on this finding, Ms. G is advised to discontinue ibuprofen therapy. Her physician is also concerned about the interval progression in Ms. G's joint stiffness and pain, and refers her to a rheumatology clinic. She reports to the rheumatologist that her pain has progressed to include both feet, both hands and wrists, both elbows, some cervical vertebrae, and the left hip. Over the past few months, she has noted difficulty with basic household tasks and has avoided physical activity. On examination, the metacarpophalangeal and proximal interphalangeal joints of both hands are found to be swollen, tender, and warm; Ms. G also has characteristic ulnar deviation of her fingers and swan-neck deformities. Skin nodules are apparent on the extensor surface of both forearms. Laboratory tests show an elevated erythrocyte sedimentation rate (ESR), low-normal hematocrit, and positive rheumatoid factor (an immune complex formed from IgM and autoreactive IgG produced in joints). Synovial fluid aspirate is notable for leukocytosis. Hand radiographs show erosion and bone loss.

Because the symptoms, examination, laboratory studies, and radiographs are consistent with a diagnosis of rheumatoid arthritis, Ms. G is started on a course of celecoxib (a COX-2 selective inhibitor), etanercept (a TNF-α antagonist), and prednisone (a glucocorticoid). Over the subsequent months, the joint pain, swelling, and tenderness decrease noticeably. Joint function in the hands is restored, and Ms. G is able to resume some physical activity.

Questions

1. Which eicosanoid mediators may be causing Ms. G's joint pains?
2. By what mechanism do glucocorticoids such as prednisone affect eicosanoid levels and/or bioactivity?
3. By what mechanism did ibuprofen cause Ms. G's gastric erosion and hemorrhage?
4. What are the potential concerns regarding long-term use of celecoxib?
5. By what mechanism does etanercept affect eicosanoid levels and/or bioactivity?

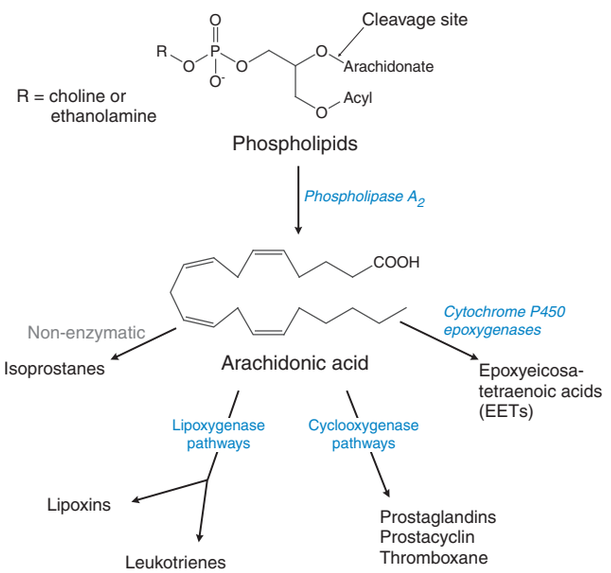


FIGURE 42-1. Overview of arachidonic acid pathways. Phospholipase A_2 acts on the phospholipids phosphatidylcholine (*PC*), phosphatidylethanolamine (*PE*), and phosphatidylinositol (*PI*) to release arachidonic acid. Unesterified arachidonic acid is then used as substrate for the cyclooxygenase, lipoxygenase, and epoxygenase pathways. The cyclooxygenase pathways produce prostaglandins, prostacyclin, and thromboxane. The lipoxygenase pathways produce leukotrienes and lipoxins. The epoxygenase pathway produces epoxyeicosatetraenoic acids (EETs). Nonenzymatic peroxidation of arachidonic acid generates isoprostanes. Phospholipase A_2 cleaves the ester bond marked by the *arrow* (at "cleavage site") to release arachidonic acid.

Generation of Arachidonic Acid and Omega-3 Fatty Acids

Arachidonic acid (cis-,cis-,cis-,cis-5,8,11,14-eicosatetraenoic acid) is the common precursor to the vast majority of eicosanoids (Fig. 42-1). Arachidonic acid must be biosynthesized from the essential fatty acid precursor **linoleic acid** (cis-,cis-9,12-octadecadienoic acid), which humans can obtain only from dietary sources. **Eicosapentaenoic acid** (cis-,cis-,cis-,cis-,cis-5,8,11,14,17-eicosapentaenoic acid; **EPA**)and **docosahexaenoic acid** (cis-,cis-,cis-,cis-,cis-,cis-4,7,10,13,16,19-docosahexaenoic acid; **DHA**) are the precursors to resolvins, protectins, and maresins. Humans can obtain EPA and DHA either from dietary sources or from biotransformation of the essential fatty acid precursor **α-linolenic acid** (cis-,cis-,cis-9,12,15-octadecatrienoic acid). α-Linolenic acid, EPA, and DHA are also called **omega-3 fatty acids** because they contain a double bond between the third and fourth carbons from the terminal (ω) end of the molecule.

In the cell, arachidonic acid does not exist as a free fatty acid, but rather is esterified to the sn_2 position of membrane phospholipids, predominantly phosphatidylcholine and phosphatidylethanolamine. Arachidonic acid is released from cellular phospholipids by the enzyme **phospholipase A_2** (Fig. 42-1), which hydrolyzes the acyl ester bond. *This important reaction, which represents the first step in the arachidonic acid cascade, is the overall rate-determining step in the generation of eicosanoids.*

There are membrane-bound and soluble isoforms of phospholipase A_2, classified as secretory ($sPLA_2$) and cytoplasmic ($cPLA_2$), respectively. The numerous phospholipase

A_2 isoforms are differentiated based on molecular weight, pH sensitivity, regulation and inhibition characteristics, calcium requirements, and substrate specificity. The existence of multiple isoforms allows for calibrated regulation of the enzyme in different tissues to achieve selective biological responses. Phospholipase A_2 isoforms relevant to inflammation are stimulated by cytokines (such as TNF-α, GM-CSF, and IFN-γ) and growth factors (such as epidermal growth factor [EGF] and the MAP kinase–protein kinase C [MAPK–PKC] cascade). Although glucocorticoids were once thought to inhibit phospholipase A_2 directly, it has now been shown that glucocorticoids act by inducing the synthesis of **lipocortins,** a family of phospholipase A_2-regulatory proteins. One of the lipocortins, annexin 1, mediates some of the anti-inflammatory actions of glucocorticoids (see below).

Cyclooxygenase Pathway

Unesterified intracellular arachidonic acid is converted by **cyclooxygenase**, **lipoxygenase**, or cytochrome-containing **epoxygenase** enzymes; the specific enzyme dictates the particular class of local eicosanoids that are generated. *The cyclooxygenase pathway leads to the formation of* ***prostaglandins***, ***prostacyclin***, *and* ***thromboxanes***; *the lipoxygenase pathways lead to* ***leukotrienes*** *and* ***lipoxins***; *and the epoxygenase pathways lead to epoxyeicosatetraenoic acids* (Fig. 42-1).

TABLE 42-1 Comparison of COX-1 and COX-2

PROPERTY	COX-1	COX-2
Expression	Constitutive	Inducible; not normally present in most tissues Constitutive in parts of nervous system
Tissue location	Ubiquitous expression	Inflamed and activated tissues
Cellular localization	Endoplasmic reticulum (ER)	ER and nuclear membrane
Substrate selectivity	Arachidonic acid, eicosapentaenoic acids	Arachidonic acid, γ-linolenate, α-linolenate, linoleate, eicosapentaenoic acids
Role	Protection and maintenance functions	Proinflammatory and mitogenic functions
Induction	Generally no induction hCG can up-regulate COX-1 in amnion	Induced by LPS, TNF-α, IL-1, IL-2, EGF, IFN-γ mRNA rises 20- to 80-fold upon induction Regulated within 1–3 hours
Inhibition	Pharmacologic: NSAIDs (low-dose aspirin)	*In vivo*: Anti-inflammatory glucocorticoids, IL-1β, IL-4, IL-10, IL-13 Pharmacologic: NSAIDs, COX-2 selective inhibitors

TABLE 42-2 Major Adverse Effects of Nonselective COX Inhibitors and COX-2 Selective Inhibitors

ADVERSE EFFECT	NONSELECTIVE COX INHIBITORS (NSAIDs)	COX-2 SELECTIVE INHIBITORS
Gastric ulceration	Yes	Yes*
Inhibit platelet function	Yes	No
Inhibit labor induction	Yes	Yes
Impair renal function	Yes	Yes
Hypersensitivity reaction	Yes	?

*The gastrointestinal toxicity of COX-2 selective inhibitors may be less than that of nonselective COX inhibitors, but there is still some incidence of toxicity.

Cyclooxygenases (also known as *prostaglandin H synthases*) are glycosylated, homodimeric, membrane-bound, heme-containing enzymes that are ubiquitous in animal cells from invertebrates to humans. *Two cyclooxygenase isoforms, denoted* ***COX-1*** *and* ***COX-2****, are found in humans.* Although COX-1 and COX-2 share ~60% sequence homology and near-superimposable three-dimensional structures, the genes are located on different chromosomes, and the enzymes differ in cellular, genetic, physiologic, pathologic, and pharmacologic profiles (Tables 42-1 and 42-2). Each cyclooxygenase catalyzes two sequential reactions. The first reaction, the cyclooxygenase step, is the oxygen-dependent cyclization of arachidonic acid to prostaglandin G_2 (PGG_2); the second reaction, the peroxidase step, is the reduction of PGG_2 to PGH_2 (Fig. 42-2).

As a result of differences in cellular localization, regulatory profile, tissue expression, and substrate requirements, COX-1 and COX-2 ultimately produce different sets of eicosanoid products that are involved in different pathways and functions. Constitutively expressed COX-1 is believed to function in physiologic, or "housekeeping," activities such as vascular homeostasis, maintenance of renal and gastrointestinal blood flow, renal function, intestinal mucosal proliferation, platelet function, and antithrombogenesis. A number of "as-needed," or specialized, functions are attributed to the products of the inducible COX-2 enzyme, including roles in inflammation, fever, pain, transduction of painful stimuli in the spinal cord, mitogenesis (particularly in the gastrointestinal epithelium), renal adaptation to stresses, deposition of trabecular bone, ovulation, placentation, and uterine contractions of labor. The role of constitutive COX-2 expression in areas of the nervous system, such as the hippocampus, hypothalamus, and amygdala, remains to be elucidated.

Protein kinetic studies suggest that there may be a third functional cyclooxygenase isoform. The putative COX-3 isoform may be a product of the same gene as COX-1 but

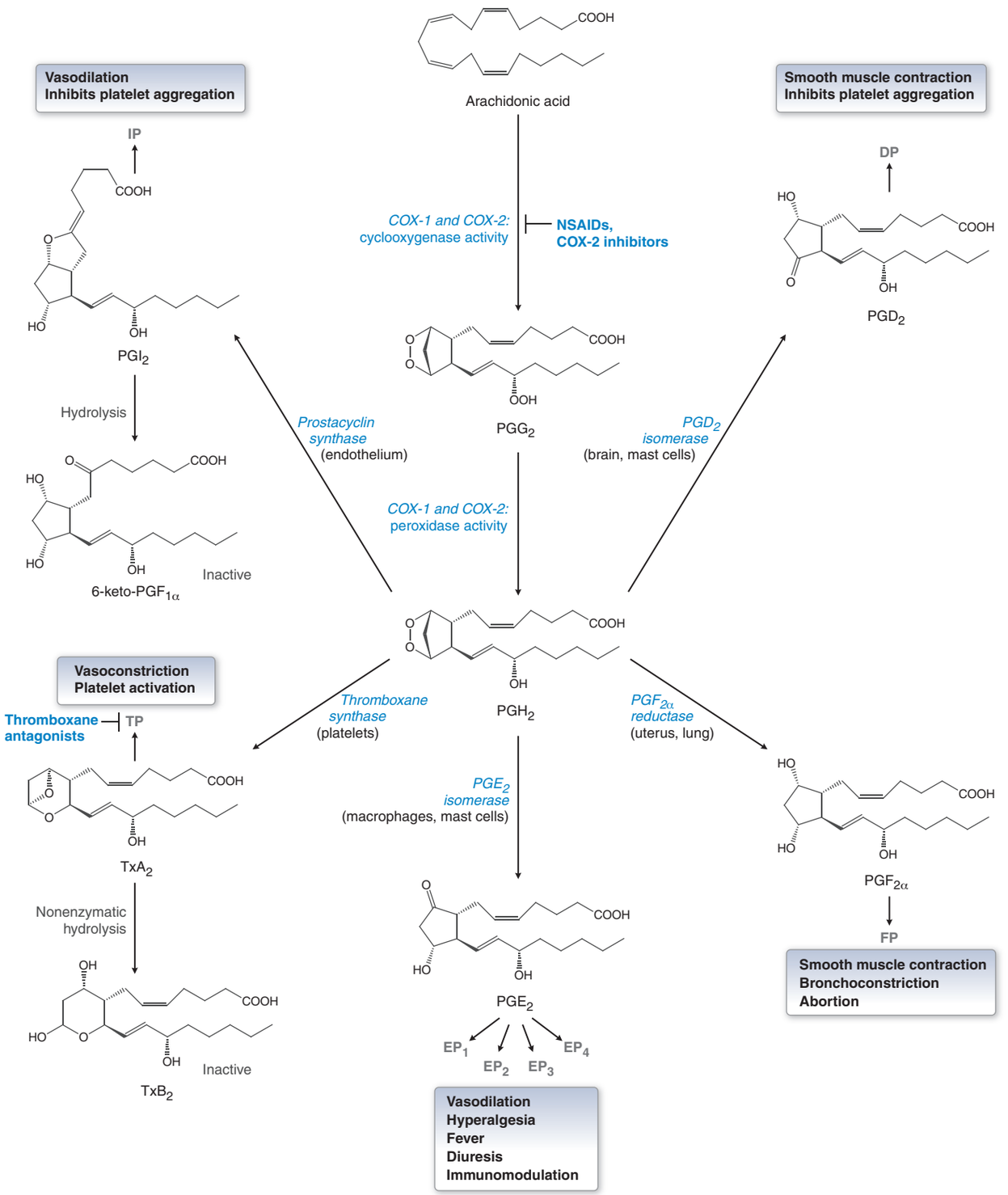
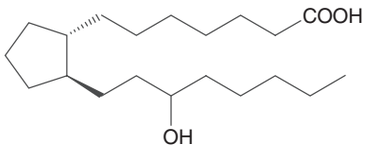


FIGURE 42-2. Prostaglandin biosynthesis, function, and pharmacologic inhibition. The biosynthetic pathways from arachidonic acid to prostaglandins, prostacyclin, and thromboxane are depicted. Tissue-specific enzyme expression determines the tissues in which the various PGH_2-derived products are biosynthesized. NSAIDs and COX-2 inhibitors are the most important classes of drugs that modulate prostaglandin production. Thromboxane antagonists and PGE_2 synthase inhibitors are promising pharmacologic strategies that are currently in development. COX, cyclooxygenase; PG, prostaglandin; Tx, thromboxane; DP, PGD_2 receptor; EP, PGE_2 receptor; FP, $PGF_{2\alpha}$ receptor; IP, PGI_2 receptor; TP, TxA_2 receptor; NSAID, nonsteroidal anti-inflammatory drug.



Prostanoid backbone

FIGURE 42-3. Prostanoid structure. The prototypical prostanoid structure is a 20-carbon carboxylic acid with a cyclopentane ring and a 15-hydroxyl group. All prostaglandins, thromboxanes, and prostacyclins are derived from this common core structure.

with different protein characteristics, possibly because of alternative mRNA splicing or post-translational modification. Furthermore, COX-3, which appears to be expressed primarily in the central nervous system, may be a potential site of action of **acetaminophen**. However, definitive proof of the existence of COX-3 remains elusive.

Prostaglandins

Prostaglandins are a large family of structurally similar compounds that each has potent and specific biological actions. The name of the family derives from their initial identification in the genitourinary system of male sheep. Prostaglandins all share a chemical structure, called a **prostanoid**, consisting of a 20-carbon carboxylic acid characterized by a cyclopentane ring and a 15-hydroxyl group (Fig. 42-3).

Prostaglandins are divided into three major subseries: PG_1, PG_2, and PG_3. The subscript numeral indicates the number of double bonds present in the molecule. The PG_2 series is the most biologically prevalent because these are the direct derivatives of arachidonic acid, which is an eicosa*tetra*enoic acid. The PG_1 series derive from the arachidonic acid precursor dihomo-γ-linolenic acid (DHGLA), an eicosa*tri*enoic acid, while the PG_3 series derive from an eicosa*penta*enoic acid (EPA, C20:5). (As alluded to earlier, protectins, D-series resolvins, and maresins derive from docosahexaenoic acid, DHA, C22:6.) The prostaglandin PGH_2 represents the critical juncture of the cyclooxygenase pathway (Fig. 42-2) because it is the immediate precursor to PGD_2, PGE_2, $PGF_{2\alpha}$, thromboxane A_2 (TxA_2), and prostacyclin (PGI_2). The distribution of these eicosanoids in various tissues is determined by the expression pattern of the different enzymes of prostaglandin synthesis (i.e., PG synthases).

The prostaglandins are important in many physiologic processes, many of which are not directly related to inflammation. These functions are highlighted in Table 42-3. Note the especially important housekeeping functions of PGE_2, broadly referred to as **cytoprotective** roles, in which organs such as gastric mucosa, myocardium, and renal parenchyma are shielded from the effects of ischemia by PGE_2-mediated vasodilation and overall regulation of blood flow. PGE_2 is also involved in inflammatory cell activation, and PGE_2 that is biosynthesized by COX-2 and PGE_2 synthase in cells near the hypothalamus appears to have a role in fever.

Thromboxane and Prostacyclin

Platelets express high levels of the enzyme thromboxane synthase but do not contain prostacyclin synthase. Therefore, *TxA_2 is the chief eicosanoid product of platelets.* TxA_2 has a half-life of only 10–20 seconds before it is nonenzymatically hydrolyzed to inactive TxB_2. TxA_2, which signals via a 7-transmembrane G protein-coupled receptor (GPCR) G_q mechanism, is a potent vasoconstrictor as well as a promoter of platelet adhesion and aggregation. In contrast, the

TABLE 42-3 Prostaglandin Products, Synthesis, Receptors, and Functions

PROSTAGLANDIN	SYNTHETIC ENZYME	TISSUES EXPRESSING SYNTHETIC ENZYME	RECEPTOR TYPE AND SIGNALING MECHANISM	FUNCTIONS
PGD_2	PGD_2 isomerase	Mast cells Neurons	DP G_s	Bronchoconstriction (asthma) Sleep control functions Alzheimer's disease
PGE_2	PGE_2 isomerase	Many tissues, including macrophages and mast cells	EP1 G_q EP2 G_s EP3 G_i EP4 G_s Other	Potentiation of responses to painful stimuli Vasodilation Bronchoconstriction Cytoprotective: modulates gastric mucosal acid secretion, mucus, and blood flow Vasodilation Bronchoconstriction Inflammatory cell activation Fever Mucus production Possibly erectile function
$PGF_{2\alpha}$	$PGF_{2\alpha}$ reductase	Vascular smooth muscle Uterine smooth muscle	FP G_q	Vascular tone Reproductive physiology (abortifacient) Bronchoconstriction

The prostanoid receptors are all G protein-coupled receptors.

vascular endothelium lacks thromboxane synthase but expresses prostacyclin synthase. Therefore, *PGI_2 is the primary eicosanoid product of the vascular endothelium.* PGI_2, which signals via G_s, functions as a vasodilator, venodilator, and inhibitor of platelet aggregation. In other words, PGI_2 is the physiologic antagonist of TxA_2. The vasodilatory actions of PGI_2, like those of PGE_2, also confer cytoprotective properties.

The local balance between TxA_2 and PGI_2 levels is critical in the regulation of systemic blood pressure and thrombogenesis. Imbalances can lead to hypertension, ischemia, thrombosis, coagulopathy, myocardial infarction, and stroke. In certain populations of the northern latitudes (including Inuit, Greenland, Irish, and Danish populations), the incidence of heart disease, stroke, and thromboembolic disorders is less than in other populations. The diet of these northern peoples is richer in fish oils and, as a result, contains relatively larger amounts of marine oils (including EPA and DHA). Analogous to the conversion of arachidonic acid into TxA_2 and PGI_2, EPA is converted into TxA_3 and PGI_3. Importantly, the vasoconstricting and platelet aggregating effects of TxA_3 are relatively weak. As a result, the thromboxane–prostacyclin balance could be tipped toward vasodilation, platelet inhibition, and antithrombogenesis. This is one possible explanation for the observation that the northern populations have a lower incidence of heart disease and is one rationale for increasing dietary fish consumption. Novel marine oil-derived mediators that possess potent anti-inflammatory and pro-resolving actions have also recently been discovered (see Lipoxins, Resolvins, Protectins, and Maresins, below).

Lipoxygenase Pathway

Besides the cyclooxygenase pathway, the other major fate of arachidonic acid is the lipoxygenase pathway, which leads to the formation of leukotrienes and lipoxins. Lipoxygenases are enzymes that catalyze the insertion of molecular oxygen (O_2) into arachidonic acid, using nonheme iron to generate specific hydroperoxides. Three lipoxygenases, 5-lipoxygenase, 12-lipoxygenase, and 15-lipoxygenase (5-LOX, etc.), are the major LOX isoforms found in humans (Table 42-4). The lipoxygenases are named for the position in arachidonic acid on which they catalyze the insertion of molecular oxygen. The immediate products of lipoxygenase reactions are hydroperoxyeicosatetraenoic acids (**HPETEs**). HPETEs can be reduced to the corresponding hydroxyeicosatetraenoic acids (**HETEs**) by glutathione peroxidase (GSP)-dependent enzymes. 5-HPETE formed by 5-LOX is the direct precursor to leukotriene A_4 (LTA_4), which itself is the precursor to all bioactive leukotrienes (Fig. 42-4). Lipoxygenases are also involved in converting 15-HETE and LTA_4 to lipoxins (Fig. 42-5).

5-LOX requires translocation to the nuclear membrane for activity. The protein 5-lipoxygenase-activating protein (**FLAP**) helps 5-LOX translocate to the nuclear membrane, form an active enzyme complex, and accept the arachidonic acid substrate from phospholipase A_2.

Leukotrienes

Leukotriene biosynthesis begins with the 5-LOX-mediated conversion of 5-HPETE to leukotriene A_4 (LTA_4). Therefore, *5-LOX catalyzes the first two steps in leukotriene biosynthesis* (Fig. 42-4). It is not known whether 5-HPETE diffuses out of the 5-LOX enzymatic active site between these steps or remains bound to the same 5-LOX enzyme for both reactions.

LTA_4 is next converted to either LTB_4 or LTC_4. The enzyme LTA_4 hydrolase converts LTA_4 to LTB_4 in neutrophils and erythrocytes. LTA_4 conversion to LTC_4 occurs in mast cells, basophils, eosinophils, and macrophages by the addition of a γ-glutamylcysteinylglycine tripeptide (glutathione). LTC_4, LTD_4, LTE_4, and LTF_4, which represent the **cysteinyl leukotrienes**, are interconverted by removal of amino acid portions of the γ-glutamylcysteinylglycine tripeptide (Fig. 42-4).

TABLE 42-4 Tissue Expression of Lipoxygenases and Products of Lipoxygenase Action

LIPOXYGENASE	TISSUE EXPRESSION	PRODUCTS	PATHWAYS	NOTES
5-LOX	Neutrophils Macrophages Mast cells Eosinophils	5-HPETE/5-HETE LTA_4 Epoxytetraene	Leukotrienes/Lipoxins Lipoxins Lipoxins/Aspirin-triggered lipoxins	Requires FLAP for activity
12-LOX				
Platelet type	Platelets Megakaryocytes (tumors)	12-HPETE/12-HETE Epoxytetraene	Lipoxins	
Epidermal type	Skin			
Leukocyte type	Macrophages GI system Brain			
15-LOX	Macrophages Monocytes Airway epithelium	15-HPETE/15-HETE Epoxytetraene	Lipoxins Lipoxins	

FLAP, 5-lipoxygenase activating protein; GI, gastrointestinal; LOX, lipoxygenase.

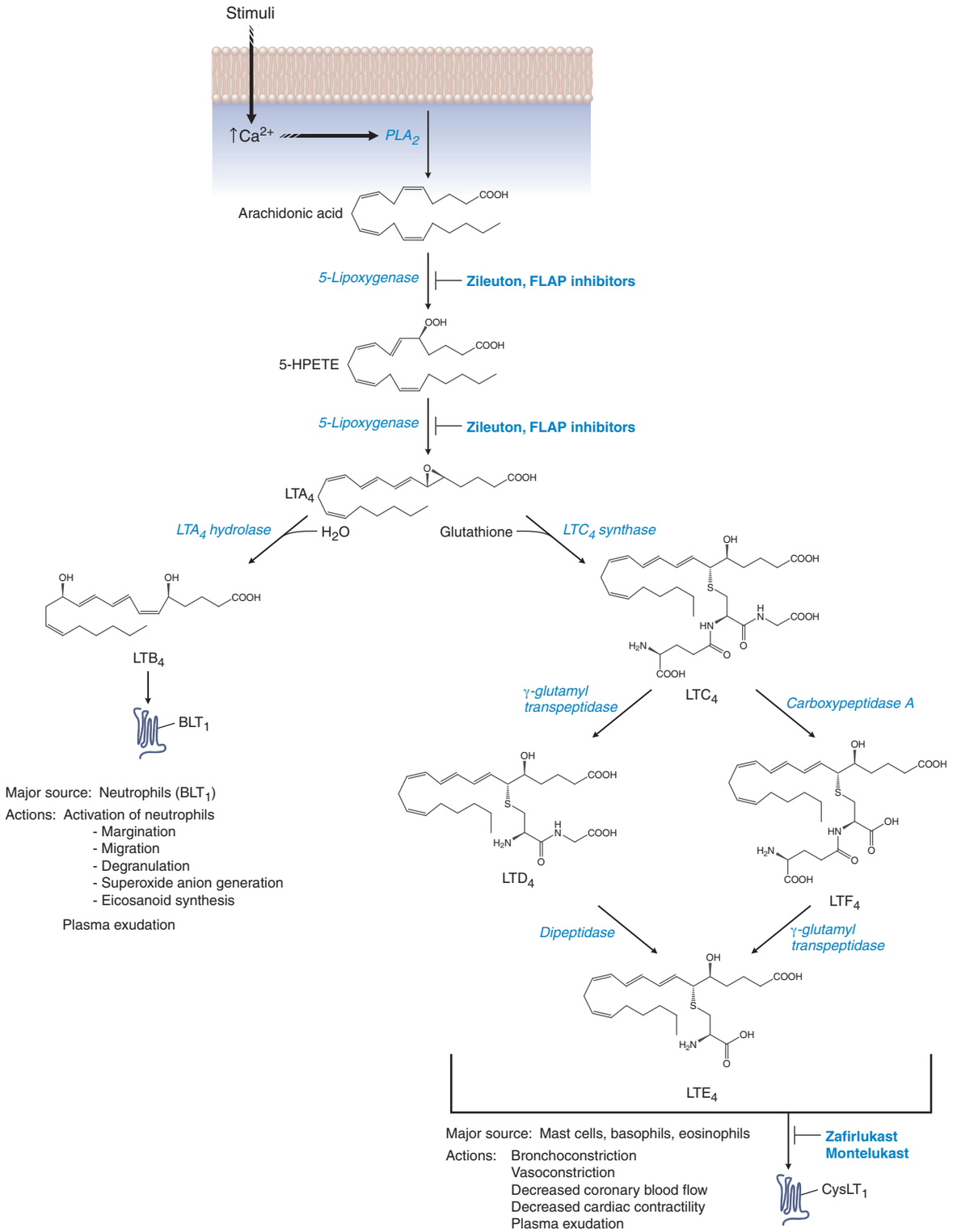


FIGURE 42-4. Leukotriene biosynthesis, function, and pharmacologic inhibition. The biosynthetic pathways from arachidonic acid to the leukotrienes are shown. Zileuton and 5-lipoxygenase activating protein (FLAP) inhibitors prevent the conversion of arachidonic acid to 5-HPETE and LTA_4; zileuton has been used in the chronic management of asthma. Zafirlukast and montelukast are antagonists at $CysLT_1$, the receptor for all cysteinyl leukotrienes (mainly LTC_4 and LTD_4); these drugs are used in the chronic management of asthma. The cysteinyl leukotrienes also interact with $CysLT_2$ (*not shown*). BLT_1 and BLT_2 are LTB_4-related G protein-coupled receptors; BLT_1 is the major LTB_4 receptor. BLT_2 is the G protein-coupled receptor for HHT, a cyclooxygenase product (see text for details; *not shown*).

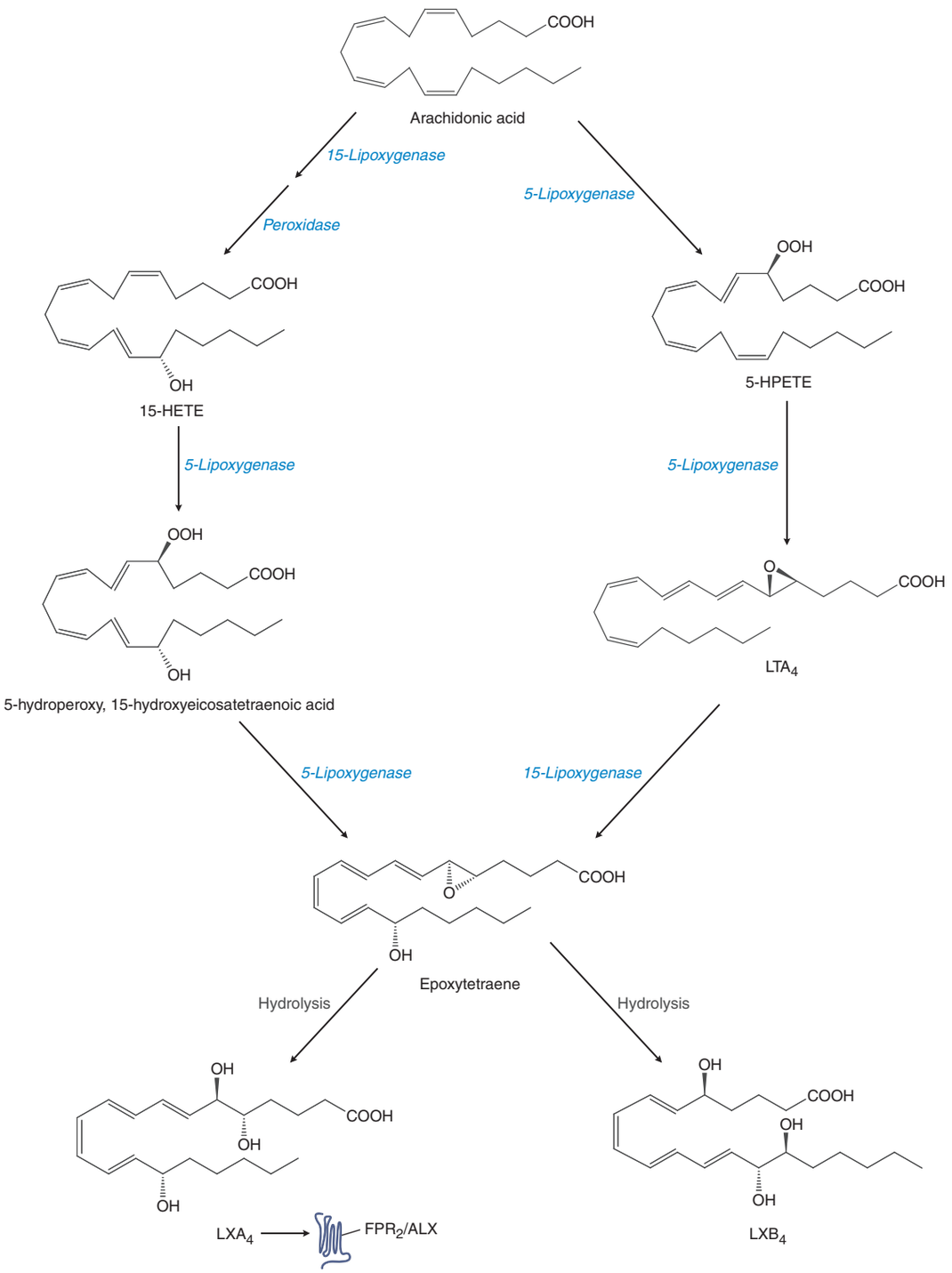


FIGURE 42-5. Lipoxin biosynthesis. Two main routes lead to biosynthesis of the lipoxins. In each pathway, sequential lipoxygenase reactions are required, followed by hydrolysis. The immediate precursor of the lipoxins is epoxytetraene; hydrolysis of epoxytetraene yields the lipoxins. **Left pathway:** Arachidonic acid is converted to 15-HETE by sequential activity of 15-lipoxygenase and peroxidase. 15-HETE is converted by 5-lipoxygenase to the chemical intermediate 5-hydroperoxy, 15-hydroxyeicosatetraenoic acid, and 5-lipoxygenase acts on this intermediate to form epoxytetraene. **Right pathway:** Arachidonic acid is converted to 5-HPETE by 5-lipoxygenase, and 5-HPETE is converted to LTA_4 by further action of 5-lipoxygenase. LTA_4 is converted to epoxytetraene by 15-lipoxygenase. **Common pathway:** Epoxytetraene is hydrolyzed to the active lipoxins LXA_4 and LXB_4. The lipoxins have both anti-inflammatory and pro-resolving roles, are counter-regulators of leukotriene action, and regulate many cytokines and growth factors. LXA_4 is a highly selective agonist for the G protein-coupled receptor FPR_2/ALX. In a transcellular reaction, platelet 12-lipoxygenase can also catalyze the formation of LXA_4 from neutrophil-derived LTA_4; the detailed mechanism remains to be elucidated (*not shown*; see also Fig. 42-7). AQ1

LTB_4 acts via two G protein-coupled receptors, BLT_1 and BLT_2. Binding of LTB_4 to BLT_1, which is expressed mainly in tissues involved in host defense and inflammation (leukocytes, thymus, spleen), leads to proinflammatory sequelae such as neutrophil chemotaxis, aggregation, and transmigration across epithelium and endothelium. LTB_4 up-regulates neutrophil lysosomal function and generates reactive oxygen species (ROS), enhances cytokine production, and potentiates the actions of natural killer (NK) cells. BLT_2 has recently been found to bind the COX product 12-HHT (12-hydroxy-5,8,10-heptadecatrienoic acid) and evoke chemotaxis of leukocytes.

The cysteinyl leukotrienes (LTC_4 and LTD_4) bind to $CysLT_1$ receptors to cause vasoconstriction, bronchospasm, and increased vascular permeability. Cysteinyl leukotrienes are responsible for the hyperreactivity to stimuli and the airway and vascular smooth muscle contraction that occur in asthmatic, allergic, and hypersensitivity processes. Together, both arms of the leukotriene pathways (i.e., LTB_4 and LTC_4/LTD_4) play key roles in psoriasis, arthritis, and various inflammatory responses. They are also key mediators in vascular disease and are likely to be important in atherosclerosis, obesity, and asthma.

Lipoxins, Resolvins, Protectins, and Maresins

Lipoxins (**lipox**ygenase **in**teraction products) are derivatives of arachidonic acid containing four conjugated double bonds and three hydroxyl groups. *The two main lipoxins, LXA_4 and LXB_4 (Fig. 42-5), modulate the actions of leukotrienes and cytokines and are important in resolution of inflammation.*

At sites of inflammation, there is typically an inverse relationship between the amounts of lipoxin and leukotriene present. This observation has led to the suggestion that lipoxins may act as counter-regulatory signals or negative regulators of leukotriene action. LXA_4 receptors are present on neutrophils and in the lung, spleen, and blood vessels. Lipoxins stop neutrophil chemotaxis, adhesion, and transmigration through endothelium (by decreasing P-selectin expression), limit eosinophil recruitment, stimulate vasodilation (by inducing synthesis of PGI_2 and PGE_2), inhibit LTC_4- and LTD_4-stimulated vasoconstriction, inhibit LTB_4 inflammatory effects, and inhibit the function of NK cells. Lipoxins stimulate the uptake and clearance of apoptotic neutrophils by macrophages and thereby mediate resolution of the inflammatory response. *Because lipoxin production appears to be important in the resolution of inflammation, an imbalance in lipoxin–leukotriene homeostasis may be a key factor in the pathogenesis of inflammatory disease.* For example, it is possible that Ms. G's chronic joint inflammation involves an imbalance in the relative amounts of leukotrienes and lipoxins in her affected joints.

Self-limited inflammatory exudates have been profiled using metabolomic approaches in murine disease models. These studies have identified the first endogenous families of omega-3-derived mediators that control both the magnitude and duration of inflammation (Fig. 42-6). The new chemical mediator families have been coined resolvins, protectins, and maresins; together with lipoxins, they constitute a chemical genus of specialized pro-resolving mediators (SPM). Resolvins, protectins, and maresins are biosynthesized from essential omega-3 fatty acid precursors, especially EPA and DHA.

The mapping of these endogenous inflammation-resolution circuits provides new avenues to probe the molecular bases of many widely occurring inflammatory diseases. Each of the resolvins, protectins, and maresins possesses multiple potent and stereoselective actions in human cells and in animal disease models. In general, these specialized local mediators limit neutrophil recruitment to sites of inflammation and stimulate macrophages to take up and remove apoptotic cells from the inflammatory site. Resolvins and protectins are produced not only in inflammatory sites but also in bone marrow and brain, where they also appear to possess potent local mediator actions. *Importantly, identification of functional SPM biosynthesized during inflammation-resolution indicates that resolution is an active process*, which is a paradigm shift away from the belief that the dampening of acute inflammation is a passive event *in vivo*. Recent findings suggest that defective or deficient resolution mechanisms may underlie some chronic inflammatory diseases and open the potential for resolution pharmacology. In the future, avenues to control inflammation may be complemented by novel therapeutics that stimulate key endogenous mechanisms of inflammation-resolution.

Epoxygenase Pathway

Microsomal cytochrome P450 epoxygenases oxygenate arachidonic acid, resulting in the formation of epoxyeicosatetraenoic acid (EET) and hydroxyacid derivatives (Fig. 42-1). The epoxygenase pathway is important in tissues that do not express COX or LOX, such as certain cells of the kidney. Epoxygenation of arachidonic acid produces four different EETs, depending on which double bond in arachidonic acid is modified. Dihydroxy derivatives of EETs, formed by hydrolysis, may regulate vascular tone by inhibiting the Na^+/K^+ ATPase in vascular smooth muscle cells and may affect renal function by regulating ion absorption and secretion. Future research may reveal more definitive functions for EETs in human physiology.

Isoprostanes

Phospholipid-esterified arachidonic acid is susceptible to free-radical-mediated peroxidation, and release of these modified lipids from the phospholipid by phospholipase A_2 gives rise to the isoprostanes (Fig. 42-1). During oxidative stress, isoprostanes are found in the blood at levels much higher than those of cyclooxygenase products. Two isoprostanes in particular, 8-epi-$PGF_{2\alpha}$ and 8-epi-PGE_2, are potent vasoconstrictors. Isoprostanes can activate NFκB, phospholipase Cγ, protein kinase C, and calcium flux. *Because the rate of formation of isoprostanes depends on cellular oxidation conditions, isoprostane levels may be indicative of oxidative stress in a wide range of pathologic conditions.* Urinary isoprostane levels are used as biomarkers of oxidative stress in ischemic syndromes, reperfusion injury, atherosclerosis, and hepatic diseases.

Metabolic Inactivation of Local Eicosanoids

Prostaglandins, leukotrienes, thromboxanes, and lipoxins are inactivated by hydroxylation, β-oxidation (resulting in a loss of two carbons), or ω-oxidation (to dicarboxylic acid derivatives). These degradation processes render the molecules more hydrophilic and excretable in the urine.

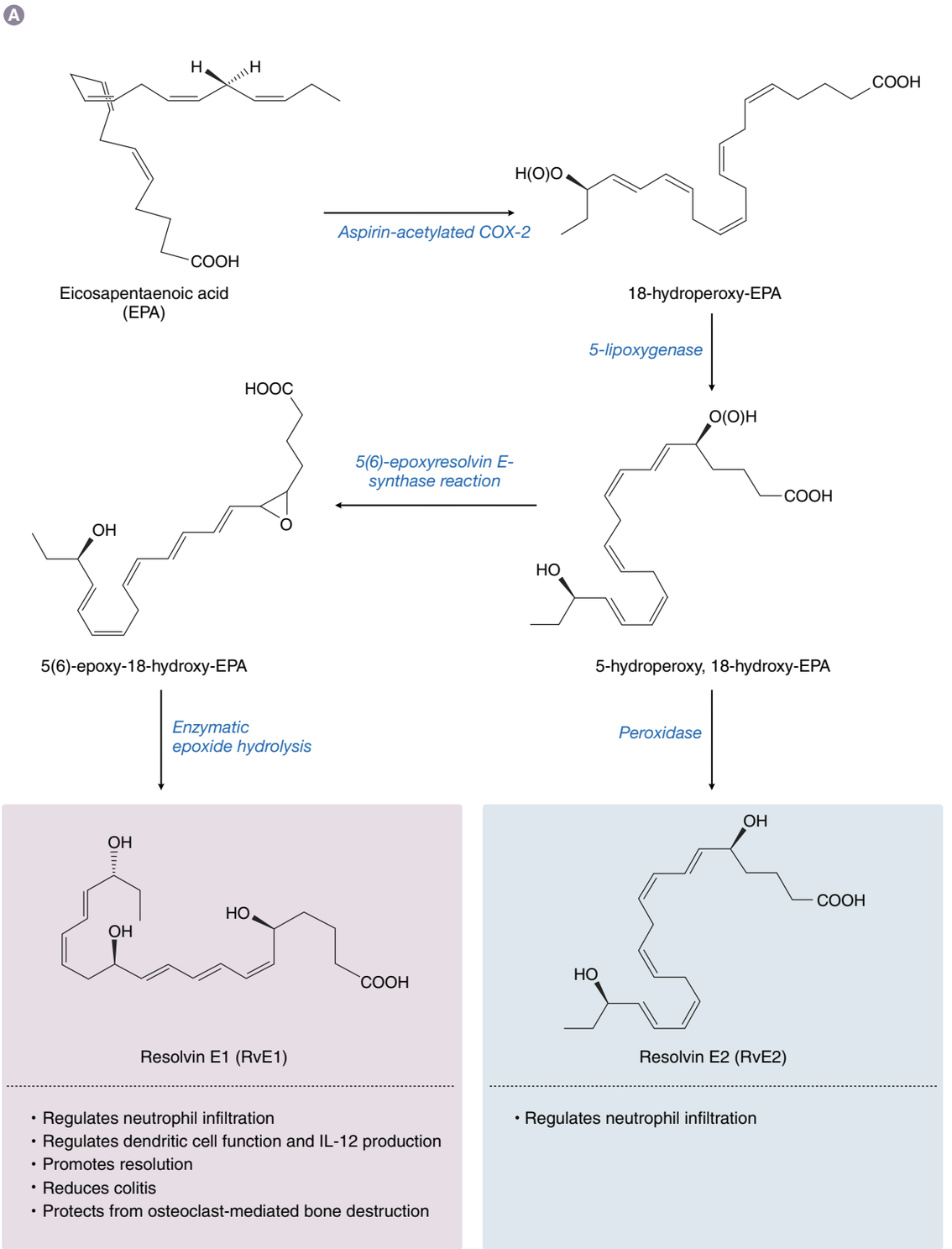


FIGURE 42-6. Resolvins, Protectins, and Maresins: Biosynthesis and Actions of Novel Families of Omega-3-Derived Mediators. A. EPA is the precursor to E-series resolvins. **B and C.** DHA is the precursor to D-series resolvins, protectins, and maresins. The major endogenous anti-inflammatory and pro-resolving functions are listed below each of the mediators. *(continued)*

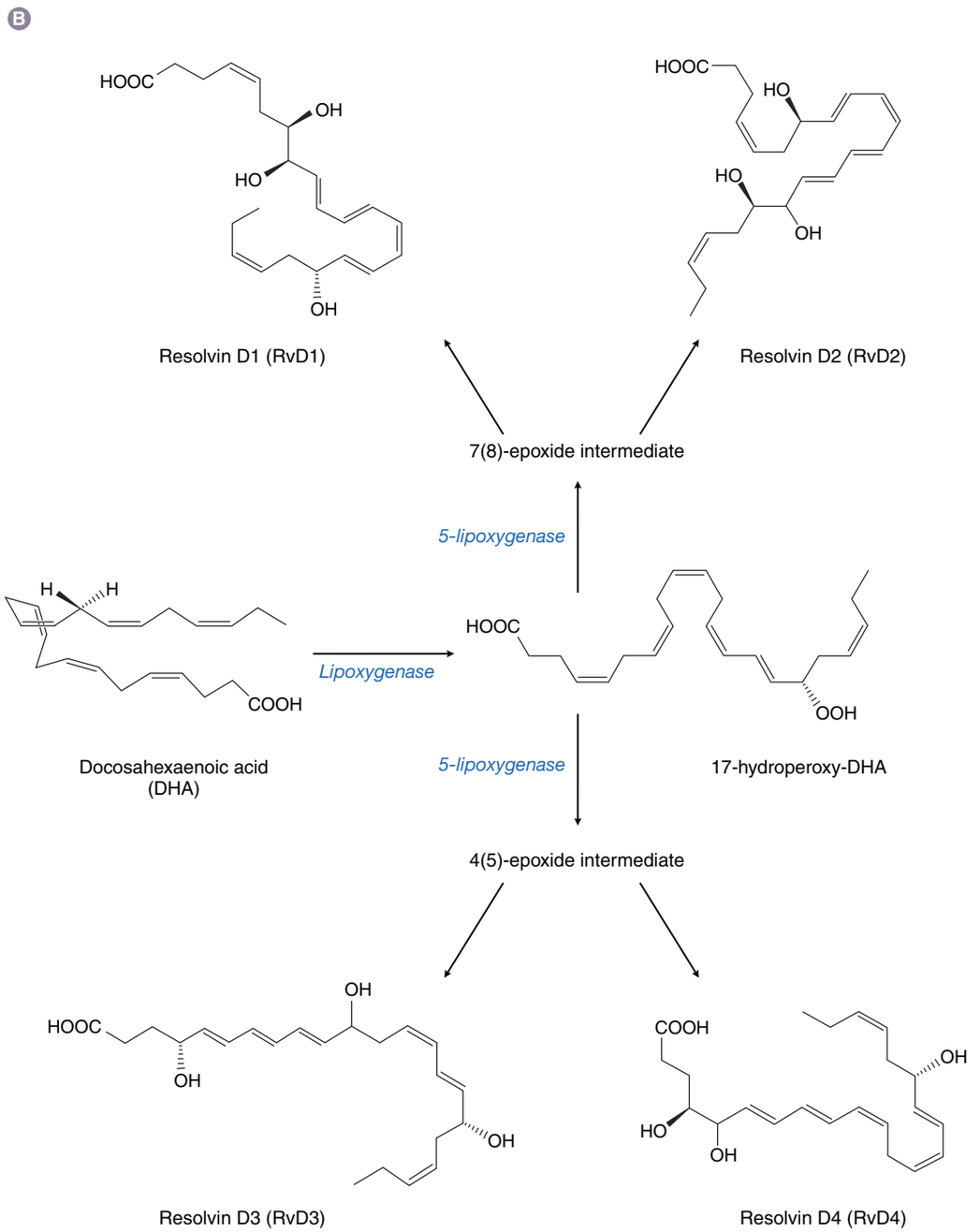


FIGURE 42-6. *(continued)*

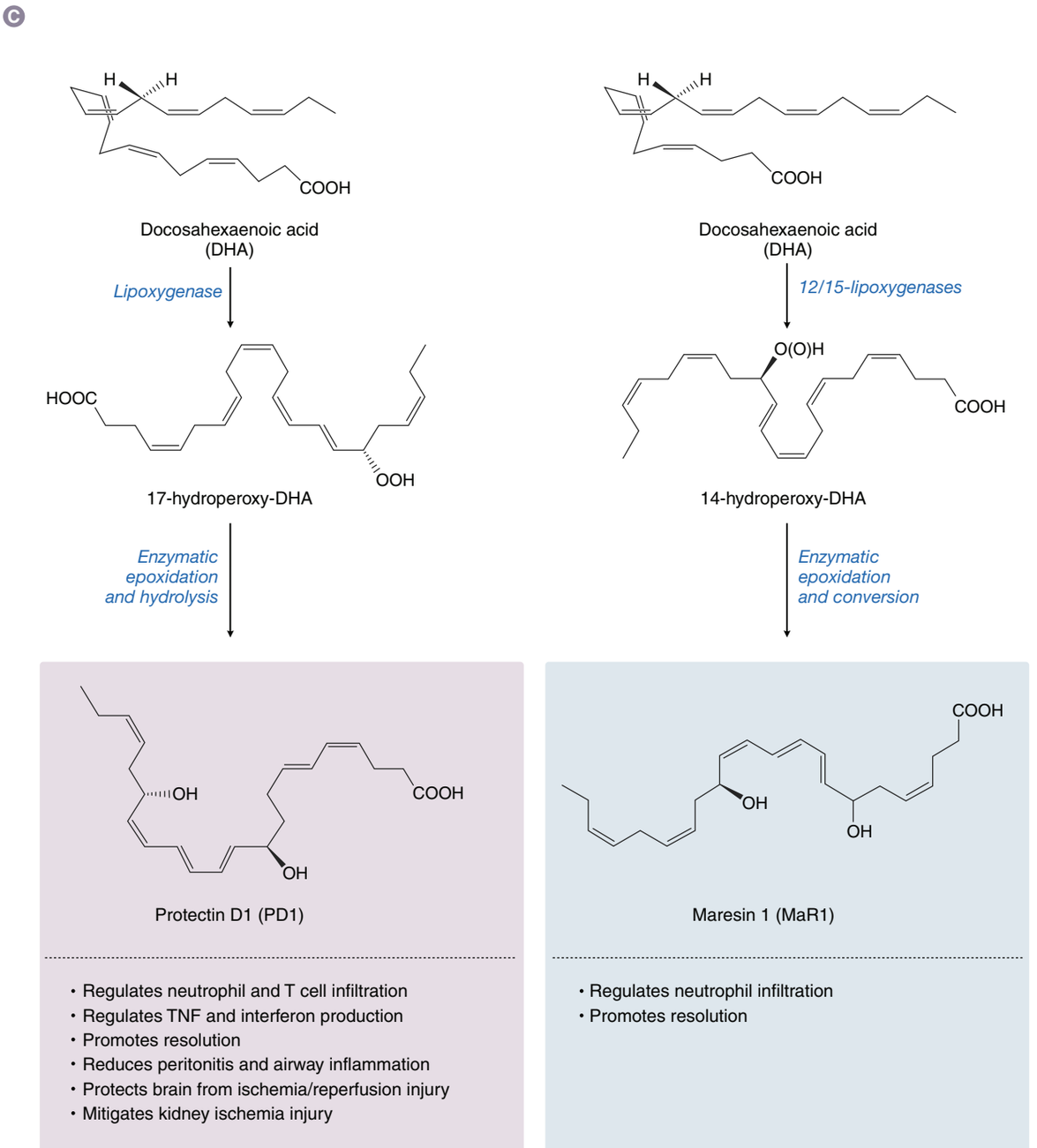


FIGURE 42-6. *(continued)*

Integrated Inflammation Schema

As described above, eicosanoids are generated locally in numerous complex reactions. It is not necessary to remember each mediator, but rather to understand the general scheme of these biosynthetic pathways. This section, along with Table 42-5, provides a concise overview of the physiologic functions of eicosanoids relevant to inflammation and host defense.

Acute inflammation is the result of an intricate network of molecular and cellular interactions induced by responses to a variety of stimuli, such as trauma, ischemia, infectious agents, or antibody reactions. Acute superficial inflammation generates local pain, edema, erythema, and warmth; inflammation in visceral organs can have similar signs and symptoms (which, in certain cases, can result in swelling against an organ capsule) and result in severe impairment of organ function.

Leukotrienes and lipoxins, as well as thromboxanes, prostaglandins, and prostacyclins, are critical for generating, maintaining, and mediating inflammatory responses. The inflammatory cascade is initiated when cells in a particular region are exposed to a foreign substance or are damaged. That insult stimulates a local cytokine cascade (including interleukins or TNF), which increases COX-2 mRNA and enzyme levels. COX-2 then facilitates production of the proinflammatory and vasoactive eicosanoids.

Locally high concentrations of PGE_2, LTB_4, and cysteinyl leukotrienes promote the accumulation and infiltration of inflammatory cells by increasing blood flow and vascular permeability. LTB_4 and 5-HETE are also important in attracting and activating neutrophils. LTB_4, biosynthesized and released by activated neutrophils at the site of inflammation, recruits and activates additional neutrophils and lymphocytes so that these cells adhere to endothelial surfaces and transmigrate into the interstitial spaces. Increased vascular permeability also results in fluid leak and cellular infiltration, causing edema.

With the aggregation of a multitude of inflammatory cells, **transcellular biosynthetic routes** are exploited to generate eicosanoids (Fig. 42-7). *In transcellular biosynthesis, eicosanoid intermediates are donated from one cell type to another to generate a larger diversity of local chemical mediators.* This demonstrates the importance of cellular adhesion and cell–cell interaction in inflammatory and immune responses.

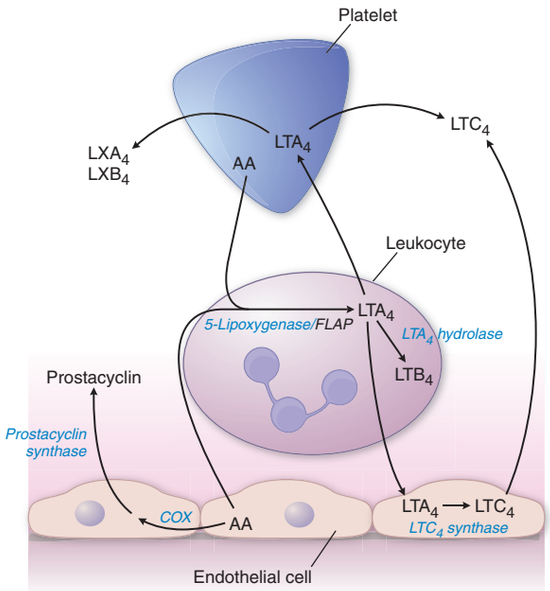


FIGURE 42-7. Examples of transcellular biosynthesis. Transcellular biosynthesis is used to generate lipoxins and cysteinyl leukotrienes locally. In the example shown here, the leukocyte (neutrophil) obtains arachidonic acid (*AA*) from platelets and uses AA to synthesize leukotriene A_4 (*LTA_4*) and leukotriene B_4 (*LTB_4*). Leukotriene A_4 is transferred from the leukocyte to platelets and endothelial cells, which synthesize and secrete leukotriene C_4 (*LTC_4*). Platelets also synthesize lipoxins (*LXA_4*, *LXB_4*) from leukotriene A_4, and endothelial cells synthesize prostacyclin using AA from endogenous stores. Note that the eicosanoids synthesized within each cell type are determined by the enzymatic repertoire of that cell type: for example, neutrophils synthesize primarily LTA_4 and LTB_4 because they express 5-lipoxygenase and LTA_4 hydrolase, whereas endothelial cells biosynthesize prostacyclin and LTC_4 because they express COX-1, COX-2, prostacyclin synthase, and LTC_4 synthase.

TABLE 42-5 Roles of Eicosanoids in the Stages of Inflammation

ACTION	EICOSANOIDS INVOLVED
Vasoconstriction	$PGF_{2\alpha}$, TxA_2, LTC_4, LTD_4, LTE_4
Vasodilation (erythema)	PGI_2, PGE_1, PGE_2, PGD_2, LXA_4, LXB_4, LTB_4
Edema (swelling)	PGE_2, LTB_4, LTC_4, LTD_4, LTE_4
Chemotaxis, leukocyte adhesion	LTB_4, HETE, LXA_4, LXB_4
Increased vascular permeability	LTC_4, LTD_4, LTE_4
Pain and hyperalgesia	PGE_2, PGI_2, LTB_4
Local heat and systemic fever	PGE_2, PGI_2, LXA_4

Feedback mechanisms are designed to ensure that the inflammatory response cannot proceed unchecked. Lipoxins help resolve inflammation and promote the return of the tissues, organs, and organism to homeostasis. COX-2-derived eicosanoids may also function in wound healing and resolution. Hence, the temporal sequence of events is critical to an organized inflammatory response. PGE_2 inhibits the functions of B and T lymphocytes and NK cells, while LTB_4 and the cysteinyl leukotrienes regulate T-cell proliferation. PGE_2 and PGI_2 are potent pain sensitizers, and lipoxins reduce nociception. These factors coordinately mediate and regulate the transition from acute to chronic inflammation (Figs. 42-2, 42-4, and 42-6).

PATHOPHYSIOLOGY OF EICOSANOIDS

Inflammation and the immune response are the body's mechanisms for combating foreign invaders. The overall scheme is designed to remove the inciting stimulus and resolve tissue damage. In some cases, the response mechanism itself causes local tissue damage, such as when activated

neutrophils inadvertently release proteases and reactive oxygen species into the local milieu. In other settings, if the inflammatory reactions persist for too long or if the immune system misidentifies a part of self as foreign, misdirected responses can cause significant and chronic tissue injury.

Profiled briefly below are selected inflammatory diseases in which eicosanoids are implicated, including asthma, inflammatory bowel disease, rheumatoid arthritis, glomerulonephritis, cancer, and cardiovascular disease. Other diseases not discussed here, but having a possible eicosanoid-related inflammatory basis, include certain skin disorders, reperfusion injuries, Alzheimer's disease, and adult respiratory distress syndrome.

Asthma

Asthma is an airway inflammatory disorder typified by intermittent attacks of dyspnea, coughing, and wheezing. Symptoms result from chronic airway inflammation, hyperreactivity, constriction, and obstruction. In asthma, antigens in the lungs stimulate cytokine cascades leading to the generation of both prostaglandins (e.g., PGD_2) and leukotrienes. Elaboration of LTB_4 attracts inflammatory cells and promotes cellular aggregation. LTB_4 acts particularly on B lymphocytes to cause cell activation, proliferation, and differentiation. LTB_4 also promotes expression of FcεRII receptors (i.e., receptors for the constant chain of IgE antibodies) on mast cells and basophils; these receptors bind IgE that is released by antigen-stimulated B lymphocytes. LTC_4 and LTD_4 are extremely potent bronchoconstrictive compounds (originally known as the slow-reacting substances of anaphylaxis, SRS-A), more than 1,000 times as potent as histamine. These cysteinyl leukotrienes also cause the airway epithelium to secrete mucus, while impairing the clearance of mucus by inhibiting the beating of cilia on airway epithelium. Mucus secretion is exacerbated by neutrophils and eosinophils, which become part of the inflammatory exudate clogging the airways. LTD_4 and LTE_4 also recruit eosinophils to asthmatic airways; eosinophils integrate signals from T lymphocytes and, when activated, release factors that damage the airway epithelium and enhance local airway inflammation.

In mouse models of asthma in which either the 5-LOX or $CysLT_1$ gene is knocked out, reduced airway hyperresponsiveness and leukocyte infiltration are observed. These results underscore the important role of leukotrienes in the pathogenesis of asthma. The role of leukotriene inhibitors in asthma treatment is discussed below; for additional information, refer to Chapter 47, Integrative Inflammation Pharmacology: Asthma.

Inflammatory Bowel Disease

Crohn's disease and ulcerative colitis are two idiopathic, chronic, relapsing, ulcerative, and inflammatory diseases of the gastrointestinal tract. Although the diseases are pathologically distinct, elevated LTB_4 production in the affected mucosa results in abnormal leukocyte infiltration into the parenchyma in both conditions. Chronic inflammation and leukocyte infiltration lead to progressive mucosal damage, with overt histologic changes. Crohn's disease is characterized by focal damage, fissuring ulcers, and granulomas, while mucosal inflammation and colonic dilatation are found in ulcerative colitis. Both diseases increase the risk of adenocarcinoma of the colon in the affected areas. Stable analogues of lipoxin A_4 are effective treatments in mouse models of Crohn's disease and bowel inflammation and may represent a promising new pharmacologic approach to the treatment of inflammatory bowel disease.

Rheumatoid Arthritis

Rheumatoid arthritis is a chronic, systemic, autoimmune, and inflammatory disease that primarily attacks the joints but also affects the skin, cardiovascular system, lungs, and muscles. Rheumatoid arthritis affects up to 1.5% of North Americans and is three times more prevalent in females than in males. Autoimmune targeting of normal joint proteins results in inflammation, with resultant local release of cytokines, TNF, growth factors, and interleukins, all of which induce COX-2 expression. *Levels of COX-2 enzyme and PGE_2 are markedly elevated in the synovial fluid of affected joints.* Other COX-2-derived eicosanoids and 5-LOX-derived leukotrienes activate the surrounding endothelium to help recruit inflammatory cells. Macrophages can release collagenase and proteases, while lymphocyte activity leads to immune complex formation; both processes further damage joint tissue and provide substrates that accelerate chronic inflammation. Common findings include synovitis, leukocytosis, rheumatoid nodules, and the presence of rheumatoid factor (a circulating antibody directed against IgG).

As a Native American woman in her fifth decade, Ms. G is in a higher risk group for rheumatoid arthritis. Autoimmune destruction of her joints resulted in the findings of a high erythrocyte sedimentation rate (consistent with a state of chronic inflammation), synovial leukocytosis, and the progressive loss of joint mobility and function. For additional information on rheumatoid arthritis, refer to Chapter 45, Pharmacology of Immunosuppression.

Glomerulonephritis

Glomerulonephritis delineates a large group of inflammatory renal conditions that ultimately lead to renal failure through deterioration of renal hemodynamics and glomerular filtration. Local complement activation promotes neutrophil and macrophage infiltration. Infiltration of the glomerulus is a characteristic early pathologic finding that correlates with abnormal levels of LTB_4, which is biosynthesized by LTA_4 hydrolase in the kidney mesangium, and which facilitates neutrophil adhesion to the glomerular mesangium and epithelium. LTA_4 is also a substrate for the biosynthesis of LTC_4 and LTD_4. All of the cysteinyl leukotrienes (LTC_4, LTD_4, LTE_4, and LTF_4) promote endothelial and mesangial proliferation. Cysteinyl leukotrienes also directly affect glomerular function; specifically, LTC_4 and LTD_4 decrease renal blood flow and glomerular filtration rate (GFR) by vasoconstricting arterioles and contracting mesangial spaces. Studies with inhibitors have confirmed the roles of leukotrienes in glomerulonephritis: LOX inhibitors administered at early stages of glomerulonephritis prevent glomerular inflammation and evidence of structural damage, and both LOX inhibitors and LTD_4 receptor antagonists increase GFR and decrease proteinuria.

Interestingly, the kidney mesangium expresses both LTA_4 hydrolase and 12-LOX, conferring the ability to synthesize either LTB_4 or LXA_4 from leukocyte-derived LTA_4. At low concentrations, LTA_4 is used primarily for LTB_4 formation;

these conditions correspond to the initiation of inflammation or to a low level of chronic inflammation. Conversely, when LTA_4 concentrations are relatively high, as in long-standing inflammation, LTA_4 is converted mostly into LXA_4, which provides an autoinhibitory, counterregulatory impact on the inflammatory response. In the glomerulus, LXA_4 counteracts the deleterious proinflammatory consequences of leukotrienes as well as the effect of leukotrienes on GFR, in part by raising afferent arteriolar flow via vasodilation.

Cancer

Long-term epidemiologic studies have suggested a correlation between chronic NSAID therapy and decreased incidence of colorectal cancer. Human colorectal adenomas and carcinomas express abundant COX-2; similar results have been found with gastric adenocarcinomas and breast tumors. In these tissues, COX-2 is believed to generate PGE_2 and other eicosanoids that promote tumor growth. The perinuclear localization of the COX-2 enzyme (Table 42-1) suggests the potential for an intracellular function of eicosanoid products in oncogenesis. Some eicosanoid derivatives can bind to homologues of the retinoic acid receptor (RXR) family of transcription factors, which are involved in many functions including the regulation of cell growth and differentiation. Overexpression of COX-2 would generate eicosanoids that could flood RXR signaling pathways and provide excessive growth stimuli. A COX-2 inhibitor is being sought as a prophylactic therapy for patients with familial adenomatous polyposis, who are at risk for colorectal cancer (see discussion of COX-2 inhibitors below).

Cardiovascular Disease

Platelet-derived thromboxane A_2 is an important mediator of thrombosis in acute coronary syndromes (unstable angina, non-ST elevation myocardial infarction, and ST elevation myocardial infarction) and other cardiovascular diseases; the COX inhibitor aspirin is an effective antiplatelet agent in the prophylaxis and treatment of these diseases (see below and Chapter 22, Pharmacology of Hemostasis and Thrombosis). Intravascular leukotriene production during the rupture of atheromatous plaques is also thought to contribute to the pathophysiology of acute coronary syndromes. Recent studies have suggested that 5-lipoxygenase, FLAP, and LTA_4 hydrolase are genetically linked to myocardial infarction, and 5-lipoxygenase inhibitors, FLAP antagonists, and LTA_4 hydrolase inhibitors may represent new classes of therapeutics for the treatment of atherosclerosis and myocardial infarction.

PHARMACOLOGIC CLASSES AND AGENTS

Pharmacologic intervention in eicosanoid biosynthesis and action is particularly useful for controlling inflammation and restoring homeostasis. Pharmacologic interventions can be directed at any of the number of steps outlined above to achieve the desired effects with tissue, spatial, and temporal selectivity. Strategies include altering the expression of key enzymes, competitively and noncompetitively inhibiting the activity of specific enzymes (e.g., PGE_2 synthase), activating receptors with exogenous receptor agonists, and preventing receptor activation with exogenous receptor antagonists. As with all aspects of medicine, therapeutic benefits must be weighed against the possible adverse effects.

Phospholipase Inhibitors

Inhibition of phospholipase A_2 prevents the release of arachidonic acid from cellular phospholipids, the rate-limiting step in eicosanoid biosynthesis. In the absence of proinflammatory mediators derived from arachidonic acid, inflammation is limited.

Glucocorticoids (also known as corticosteroids, including **prednisone**, **prednisolone**, and **dexamethasone**) are a mainstay of therapy in a multitude of autoimmune and inflammatory diseases. Glucocorticoids induce a family of secreted calcium- and phospholipid-dependent proteins called **lipocortins**. Lipocortins interfere with the action of phospholipase A_2 and thereby limit the availability of arachidonic acid. Annexins, such as annexin 1 and annexin 1-derived peptides, are also induced by glucocorticoids. In turn, annexins act at G protein-coupled receptors on leukocytes to block proinflammatory responses and enhance endogenous anti-inflammatory mechanisms; one anti-inflammatory mechanism involves activation of the lipoxin A_4 receptor. Glucocorticoids also inhibit the action of COX-2 and the formation of prostaglandins by several mechanisms: (1) repressing COX-2 gene and enzyme expression; (2) repressing the expression of cytokines that activate COX-2; and (3) as noted above, limiting the available pool of COX-2 substrate (arachidonic acid) by indirectly blocking phospholipase A_2. Because of this profound and global suppression of immune and inflammatory responses, glucocorticoids are used to treat a number of autoimmune conditions (see Chapter 45, Pharmacology of Immunosuppression).

Small-molecule inhibitors of specific phospholipases are in development; these compounds may offer the potential for decreased adverse effects compared to those associated with glucocorticoid use. See Chapter 28, Pharmacology of the Adrenal Cortex, for a more extensive discussion of the effects of glucocorticoids.

Cyclooxygenase Inhibitors

Cyclooxygenase pathway inhibitors are some of the most frequently prescribed drugs in medicine. The nonsteroidal anti-inflammatory drugs (NSAIDs) and acetaminophen are the most commonly used agents in this class.

Traditional Nonselective Inhibitors: NSAIDs

NSAIDs are important because of their combined anti-inflammatory, antipyretic, and analgesic properties. The ultimate goal of most NSAID therapies is to inhibit the COX-mediated generation of proinflammatory eicosanoids and to limit the extent of inflammation, fever, and pain. The drugs' antipyretic activity is likely related to their ability to decrease the levels of PGE_2, particularly in the region of the brain surrounding the hypothalamus. *Despite the benefits of current NSAIDs, these drugs only suppress the signs of the underlying inflammatory response but may not necessarily reverse or resolve the inflammatory process.*

A multitude of NSAIDs have been developed over the last century; most are polycyclic carboxylic acid derivatives. Except for aspirin, all NSAIDs act as reversible, competitive inhibitors of cyclooxygenase (Fig. 42-2). These drugs block the hydrophobic channel of the cyclooxygenase protein where the substrate arachidonic acid binds, thereby preventing conversion of arachidonic acid to PGG_2. Traditional NSAIDs inhibit both COX-1 and COX-2 to different

degrees. Because of inhibition of COX-1, long-term NSAID therapy has many deleterious effects. The cytoprotective roles of the COX-1 eicosanoid products are eliminated, leading to a spectrum of **NSAID-induced gastropathy** including dyspepsia, gastrotoxicity, subepithelial damage and hemorrhage, gastric mucosal erosion, frank ulceration, and gastric mucosal necrosis. As in Ms. G's case, patients with gastric ulceration can have bleeding into the stomach, where reaction of hemoglobin with stomach acid results in hematemesis, or vomiting of material that, when regurgitated, has the color and consistency of coffee grounds. Regulation of blood flow to the kidney is likewise perturbed, decreasing GFR and potentially causing renal ischemia, papillary necrosis, interstitial nephritis, and renal failure. The results of studies on the effects of COX-2 inhibitors (see below) have prompted reinvestigation of the effects of COX-1 inhibitors and traditional NSAIDs, with the finding that these classes of drugs may also be associated with cardiac risks. The Food and Drug Administration (FDA) has asked nonprescription NSAID manufacturers to update labeling with specific information about cardiovascular risks and to "remind patients of the limited dose and duration of treatment of these products." Epidemiologic studies suggest that 20–30% of hospitalizations of patients over the age of 60 are due to complications of NSAID use.

The organic acid functionality of NSAIDs confers important pharmacokinetic properties on these drugs, including near-complete absorption from the gut, binding to plasma albumin, accumulation in cells at the site of inflammation, and efficient renal excretion. NSAIDs can be divided into short (<6 hours) and long (>10 hours) half-life classes. NSAIDs with long elimination half-lives include **naproxen**, **salicylate**, **piroxicam**, and **phenylbutazone**.

Chemical classification of the NSAIDs is based on the structure of a key moiety in each subclass of drugs (Fig. 42-8). The following discussion groups the NSAIDs by chemical class; a discussion of the choice of a particular NSAID for a given clinical situation follows the descriptions of the individual agents.

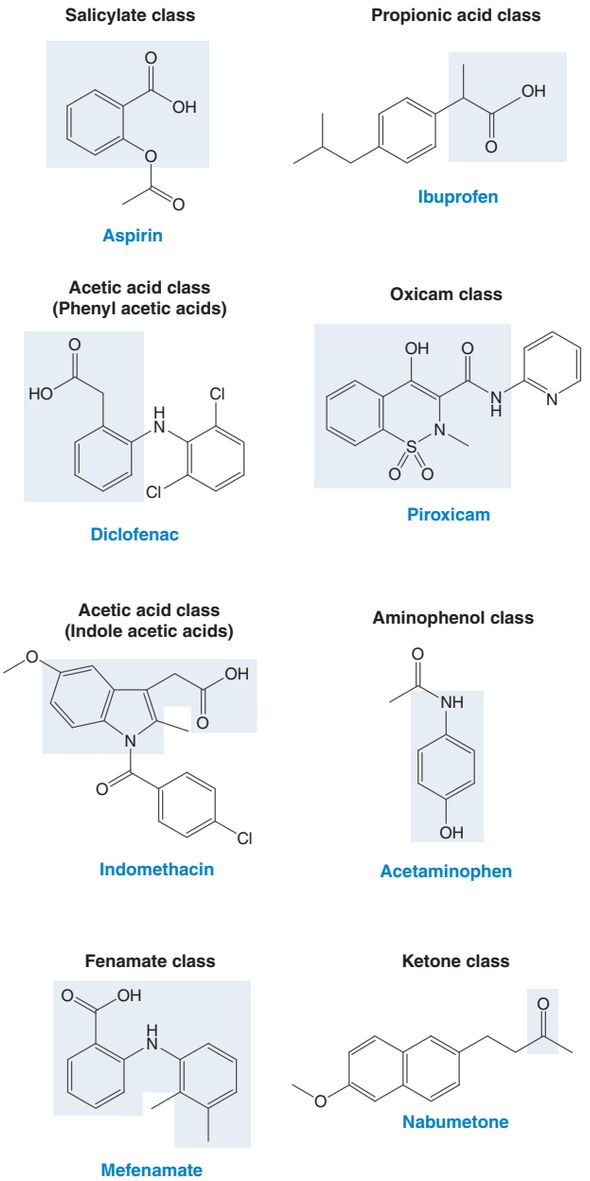


FIGURE 42-8. Structural classes of NSAIDs. NSAIDs are generally hydrophobic molecules, most of which have a carboxylic acid group. NSAIDs are categorized by class depending on one or more of the key moieties in the structure. The moiety that is common to members of each class is highlighted by a box. The structure helps to determine the pharmacokinetic properties of each particular NSAID. Note that acetaminophen is not actually an NSAID, because it has only weak anti-inflammatory properties; this drug is included here because, like NSAIDs, acetaminophen is commonly used for its analgesic and antipyretic effects.

Salicylates

Salicylates include **aspirin** (acetylsalicylic acid) and its derivatives. Aspirin is the oldest of the NSAIDs and is widely used to treat mild-to-moderate pain, headache, myalgia, and arthralgia. *In contrast to other NSAIDs, aspirin acts in an irreversible manner by acetylating the active-site serine residue in both COX-1 and COX-2.* Acetylation of COX-1 destroys the enzyme's cyclooxygenase activity, preventing the formation of COX-1-derived prostaglandins, thromboxanes, and prostacyclin. Salicylates (along with indomethacin, piroxicam, and ibuprofen) may also inhibit the neutrophil oxidative burst by reducing NADPH oxidase activity.

Daily low-dose aspirin is used as an antithrombogenic agent for both prophylaxis and postevent management of acute coronary syndromes and ischemic stroke. Recall that aspirin is antithrombogenic because of its irreversible inhibition of COX, which prevents platelets from biosynthesizing TxA_2. Within an hour of oral aspirin administration, the COX-1 activity in existing platelets is irreversibly destroyed. Platelets, lacking nuclei, cannot synthesize new protein. Therefore, the irreversibly acetylated COX-1 enzymes cannot be replaced by freshly synthesized proteins, and the platelets' cyclooxygenase activity is irreversibly inhibited for their circulating lifetime (about 10 days). Although aspirin also irreversibly inhibits vascular endothelial cell COX-1 and COX-2, the endothelial cells can synthesize new COX protein and thus can rapidly resume synthesis of PGI_2. *A single administration of aspirin thus decreases for several days the amount of thromboxane that can be generated, shifting the vascular TxA_2-PGI_2 balance toward PGI_2-mediated vasodilation, platelet inhibition, and antithrombogenesis.*

Aspirin-mediated inhibition of COX-2 prevents the generation of prostaglandins. Unlike COX-1, which is totally

inactivated by aspirin, the aspirin-modified COX-2 enzyme retains a distinct part of its catalytic activity and can form a new product, 15-(R)-HETE, from arachidonic acid. By analogy to lipoxin biosynthesis (Fig. 42-5), 5-LOX then converts 15-(R)-HETE to 15-epi-lipoxins, which are relatively stable stereoisomers (carbon 15-position epimers) of lipoxins that are collectively called **aspirin-triggered lipoxins (ATLs)**. *15-Epi-lipoxins mimic the functions of lipoxins as anti-inflammatory agents.* 15-Epi-lipoxins may represent another endogenous mechanism of anti-inflammation, and their production mediates at least part of the anti-inflammatory effects of aspirin. Development of 15-epi-lipoxin analogues could lead to anti-inflammatory drugs that do not have the adverse effects associated with COX-1 inhibition.

Aspirin is generally well tolerated. Its major toxicities are the gastropathy and nephropathy common to all NSAIDs. Long-term aspirin therapy can lead to gastrointestinal ulceration and hemorrhage, nephrotoxicity, and hepatic injury. NSAIDs should be used cautiously, if at all, in patients with renal insufficiency and heart failure. Two unique toxicities are **aspirin-induced airway hyperreactivity** in asthmatics (so-called *aspirin-sensitive asthma*) and **Reye's syndrome**. The prevalence of aspirin sensitivity among patients with asthma is approximately 10%. Exposure to aspirin in these patients leads to ocular and nasal congestion along with severe airway obstruction. Aspirin-sensitive patients are also reactive to some other NSAIDs, including indomethacin, naproxen, ibuprofen, mefenamate, and phenylbutazone. One possible etiology of aspirin/NSAID sensitivity in asthmatics is that exposure to these drugs leads to increased levels of leukotrienes, which are implicated in the pathogenesis of asthma (see Fig. 42-1).

Reye's syndrome is a condition characterized by hepatic encephalopathy and liver steatosis in young children. Aspirin therapy during the course of a febrile viral infection has been implicated as a potential etiology of the liver damage. Although a causal link between aspirin and Reye's syndrome has not been definitively established, aspirin is generally not administered to children because of the fear of Reye's syndrome. Acetaminophen is widely used in children instead of aspirin.

Propionic Acid Derivatives

Propionic acid NSAIDs include **ibuprofen**, **naproxen**, **ketoprofen**, and **flurbiprofen**. Ibuprofen is a relatively potent analgesic used in rheumatoid arthritis (as in the case of Ms. G, to relieve intermittent pain), osteoarthritis, ankylosing spondylitis, gout, and primary dysmenorrhea. Naproxen has a long plasma half-life, is 20 times more potent than aspirin, directly inhibits leukocyte function, and causes less severe gastrointestinal adverse effects than aspirin.

Acetic Acid Derivatives

Acetic acid NSAIDs include the indole acetic acids—**indomethacin**, **sulindac**, and **etodolac**—and the phenylacetic acids **diclofenac** and **ketorolac** (a substituted phenylacetic acid derivative). Besides inhibiting cyclooxygenase, many of the acetic acid NSAIDs promote the incorporation of unesterified arachidonic acid into triglyceride, thus reducing the availability of the substrate for cyclooxygenase and lipoxygenase. Indomethacin is a direct inhibitor of neutrophil motility, but it is not tolerated by patients as well as ibuprofen. Diclofenac also reduces intracellular arachidonic acid concentrations by altering cellular fatty acid transport. Diclofenac is a more potent anti-inflammatory than indomethacin and naproxen and is used widely in the treatment of pain associated with renal stones. Ketorolac is primarily employed for its strong analgesic properties, particularly in postsurgical patients; however, in part due to its potency and adverse effects, ketorolac is used for no more than 3–5 days.

The acetic acid NSAIDs are mostly used to relieve symptoms in the long-term treatment of rheumatoid arthritis, osteoarthritis, ankylosing spondylitis, and other musculoskeletal disorders. Use of acetic acid NSAIDs causes gastrointestinal ulceration and, rarely, hepatitis and jaundice. Indomethacin also has a specific use in promoting the closure of a patent ductus arteriosus in newborns by inhibiting the vasodilatory eicosanoids PGE_2 and PGI_2.

Oxicam Derivatives

Piroxicam is as efficacious as aspirin, naproxen, and ibuprofen in the treatment of rheumatoid arthritis and osteoarthritis, but may be better tolerated. Piroxicam has additional effects in the modulation of neutrophil function by inhibiting collagenase, proteoglycanase, and the oxidative burst. Because of its extremely long half-life, piroxicam can be administered once daily. As with other NSAIDs, piroxicam displays gastrointestinal adverse effects such as gastric ulceration, and it prolongs the bleeding time because of its antiplatelet effect.

Fenamate Derivatives

The two fenamate NSAIDs are **mefenamate** and **meclofenamate**. Both inhibit cyclooxygenases but also antagonize prostanoid receptors to various degrees. Because fenamates have less anti-inflammatory activity and are more toxic than aspirin, there is little advantage to their use. Mefenamate is used only for primary dysmenorrhea, and meclofenamate is used in the treatment of rheumatoid arthritis and osteoarthritis.

Ketone NSAIDs

Nabumetone is a ketone prodrug that is oxidized *in vivo* to the active acid form. Compared to other nonselective NSAIDs, nabumetone has preferential activity against COX-2. The incidence of gastrointestinal adverse effects is relatively low, although headache and dizziness are frequently reported.

Acetaminophen

Acetaminophen is sometimes classified with the NSAIDs, but it is technically not an NSAID: *although acetaminophen has analgesic and antipyretic effects similar to aspirin, the anti-inflammatory effect of acetaminophen is insignificant because of its weak inhibition of cyclooxygenases.* Nonetheless, acetaminophen therapy can be valuable in patients, such as children, who are at risk for the adverse effects of aspirin. The most important adverse effect of acetaminophen is hepatotoxicity. Modification of acetaminophen by hepatic cytochrome P450 enzymes produces a reactive metabolite, which is normally detoxified by conjugation with glutathione. An overdose of acetaminophen can overwhelm glutathione stores, leading to cellular and oxidative damage and, in severe cases, to acute hepatic necrosis (see Chapter 5, Drug Toxicity).

Selection of the Appropriate NSAID

The anti-inflammatory, analgesic, and antipyretic effects of the NSAIDs appear to vary among the many agents in

this class. However, despite differences in chemistry, tissue selectivity, enzyme selectivity, pharmacokinetics, and pharmacodynamics, the differences in efficacy may not be clinically significant. Overall, the rationale and choice of NSAID do not generally make a substantial difference in treating rheumatoid arthritis or osteoarthritis. However, successful NSAID therapy is still considered more of an art than a science, and therapy for each patient should be directed at achieving the desired anti-inflammatory, analgesic, and antipyretic effects while minimizing adverse effects. The adverse gastric effects of long-term NSAID therapy can be reduced by co-administration of histamine H_2 receptor antagonists or proton pump inhibitors (refer to Chapter 46, Integrative Inflammation Pharmacology: Peptic Ulcer Disease).

COX-2 Inhibitors

As noted above, long-term NSAID therapy can be associated with severe gastrointestinal adverse effects that are thought to be caused by inhibition of gastrointestinal COX-1. It was hypothesized that selective inhibition of COX-2 could have the theoretical advantage of inhibiting the chemical mediators responsible for inflammation while maintaining the cytoprotective effects of the products of COX-1 activity.

COX-2 Selective Inhibitors

Although COX-2 was identified only in the 1990s, intense research swiftly led to the development of COX-2 selective inhibitors for clinical use. *Compared with COX-1, COX-2 has a larger hydrophobic channel through which substrate (arachidonic acid) enters the active site.* Subtle structural differences between COX-2 and COX-1 allowed the development of drugs that act preferentially on COX-2.

The COX-2 selective inhibitors—**celecoxib**, **rofecoxib**, **valdecoxib**, and **meloxicam** (Fig. 42-9)—are sulfonic acid derivatives that exhibit 100 times greater selectivity for COX-2 than for COX-l. The relative inhibition of the two cyclooxygenase isozymes in any given tissue is also a function of drug metabolism, pharmacokinetics, and possibly enzyme polymorphisms. The COX-2 selective inhibitors have anti-inflammatory, antipyretic, and analgesic properties similar to the traditional NSAIDs, but they do not share the antiplatelet actions of the COX-1 inhibitors. Various coxibs had been approved for treatment of osteoarthritis, rheumatoid arthritis, acute pain in adults, and primary dysmenorrhea. However, relative to other NSAIDs, the safety profile of the COX-2 selective inhibitors is uncertain. At the present time, only celecoxib is an approved drug in the United States. Rofecoxib was withdrawn from the worldwide market in 2004 because of an increase in thrombogenicity with prolonged use, manifesting as myocardial infarction and stroke; valdecoxib was then withdrawn in 2005.

The increased thrombogenicity uncovered in clinical use may be due to prolonged inhibition of vascular COX-2 within endothelial cells, leading to reduced PGI_2 formation. In addition, inhibition of COX-2 may generate problems in wound healing, angiogenesis, and the resolution of inflammation. COX-2 selective inhibitors are much more expensive than equivalent doses of many NSAIDs, especially aspirin and indomethacin. Note that Ms. G's physician attempted to take advantage of the relative gastrointestinal safety of COX-2 selective inhibitors when she was switched from ibuprofen to a COX-2 inhibitor, in part because of symptomatic and endoscopic evidence of NSAID-induced

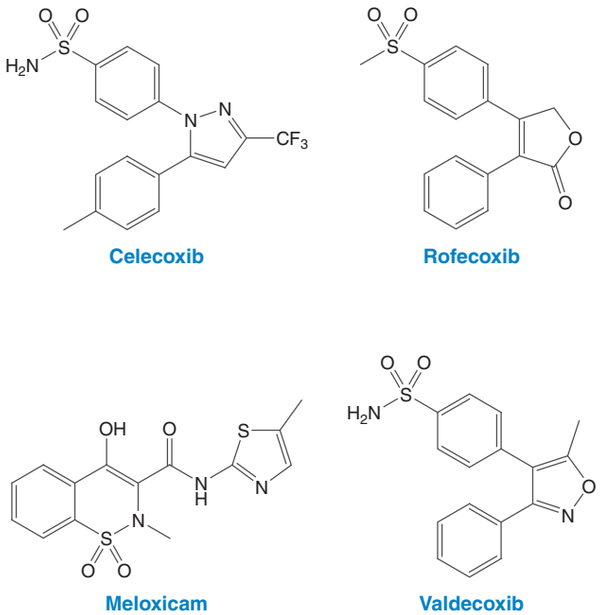


FIGURE 42-9. COX-2 selective inhibitors. COX-2 selective inhibitors are hydrophobic sulfonic acid derivatives. Like traditional NSAIDs, these molecules block the hydrophobic channel leading to the active site of cyclooxygenase and thus inhibit the enzyme. Note that COX-2 selective inhibitors are generally larger molecules than NSAIDs. These drugs preferentially inhibit COX-2 compared to COX-1 because the hydrophobic channel of COX-2 is larger than that of COX-1. (That is, COX-2 selective inhibitors are too bulky to access the smaller hydrophobic channel of the COX-1 enzyme.) The COX-2 selective inhibitors display approximately 100-fold greater selectivity for COX-2 compared to COX-1.

gastropathy. It is being increasingly recognized, however, that the COX-2 inhibitors may not have as significant an advantage over traditional NSAIDs as previously thought in reducing gastropathy and gastrointestinal bleeding. For example, one study with rofecoxib demonstrated a fivefold increase in significant upper gastrointestinal bleed compared to placebo. One possible mechanism for this toxicity could be the adverse effect of COX-2 inhibitors on the healing of gastric ulcerations.

Celecoxib remains the only FDA-approved COX-2 selective inhibitor. Currently approved indications include osteoarthritis, rheumatoid arthritis, juvenile rheumatoid arthritis (>2 years of age), ankylosing spondylitis, acute pain in adults, and primary dysmenorrhea. Celecoxib is also approved to reduce the number of adenomatous colorectal polyps in individuals with **familial adenomatous polyposis**. Celecoxib decreases the activity of peroxisome proliferator activated receptor δ (PPARδ), a transcription factor that heterodimerizes with the RXR transcription factors involved in growth regulation. It is not yet clear whether COX-2 inhibitors bind directly to PPARδ or whether they lead indirectly to the production of other molecules that inhibit PPARδ. In either case, inhibition of PPARδ prevents signaling through the PPARδ pathway and thus removes a potent mitogenic stimulus that could promote the development of colon cancer.

Like other coxibs, celecoxib carries a label warning of increased, possibly fatal, and possibly dose- and duration-dependent cardiovascular thrombotic events (myocardial infarction and stroke). Celecoxib also increases the

risks of hypertension, edema, and heart failure, particularly at higher doses. Celecoxib is contraindicated in the treatment of pain associated with coronary artery bypass surgery.

A primary consideration in prescribing analgesic therapy with a coxib is whether the patient has a concurrent need for an anti-inflammatory agent. If the patient requires primarily analgesia, then acetaminophen may suffice, perhaps in combination with adjunct analgesics or adjunct therapies (e.g., for arthropathies, consider physical therapy or surgical intervention). If, however, there is an established indication for chronic anti-inflammatory therapy and there is also a risk factor for gastropathy (e.g., history of ulcer disease, elderly, concurrent antiplatelet or anticoagulant or glucocorticoid therapy), then a coxib or a combination regimen with an NSAID and a proton pump inhibitor may be considered. In all cases, the risks of coxibs in patients at risk for ischemic heart disease and cerebrovascular disease must be considered as part of the overall risk/benefit analysis.

It was hoped that the second-generation COX-2 inhibitors in development—such as parecoxib (a water-soluble prodrug of valdecoxib), etoricoxib, and lumiracoxib—would demonstrate increased selectivity for COX-2 over COX-1 and would not have the adverse cardiovascular effects of the existing COX-2 inhibitors. However, further clinical development of this class of drugs is in question.

Cytokine Inhibitors

The proinflammatory cytokines TNF-α and IL-1 enhance prostaglandin production and up-regulate COX-2. Novel molecular technologies have provided the ability to inhibit the action of these cytokines and thus to inhibit the process whereby an injurious stimulus activates COX-2 and initiates the inflammatory response. Four antibody-based TNF-α antagonists are currently available: **etanercept**, **infliximab**, **adalimumab**, and **golimumab**. Etanercept consists of the extracellular domain of the TNF-α receptor coupled to human IgG1; infliximab is a humanized mouse monoclonal antibody directed against TNF-α; and adalimumab and golimumab are humanized monoclonal IgG1 antibodies directed against TNF-α.

TNF-α antagonists were first approved for treatment of rheumatoid arthritis. With few adverse effects, these drugs halt joint destruction and bone erosion, decrease pain, calm swollen and tender joints, and limit overall disease progression in rheumatoid arthritis. Anti-TNF antibody drugs have also been approved for use in a variety of other autoimmune diseases (see Chapter 45) such as ankylosing spondylitis, psoriatic arthritis, plaque psoriasis, juvenile idiopathic arthritis (>4 years of age for adalimumab and >2 years of age for etanercept), Crohn's disease (adalimumab and infliximab), and ulcerative colitis (infliximab). Years of experience with this class of drugs has shown an increased risk of serious infections, including disseminated or extrapulmonary tuberculosis, invasive fungal infections (*Aspergillus* and endemic fungi such as *Histoplasma*), hepatitis B virus reactivation, and opportunistic infections. Patients are routinely tested for latent tuberculosis prior to initiation of therapy and must be monitored for active tuberculosis during treatment. Other adverse effects include a small but possibly increased risk of lymphoma, demyelinating disease, heart failure, and pancytopenia.

Lipoxins, ATLs, and lipoxin-stable analogues also block the actions of TNF-α, providing a potential new treatment approach (see below).

Anakinra is a recombinant form of the human IL-1 receptor produced in *Escherichia coli*; this drug is approved for use in patients with rheumatoid arthritis who have failed one or more disease-modifying antirheumatic agents. Analogous to TNF antagonists, anakinra is associated with an increased risk of serious infection. Additional IL-1 antagonists are being developed for use in inflammatory and autoimmune diseases. For more information on these agents, refer to Chapter 45.

Prostanoid Receptor Mimetics

Several applications for prostanoid receptor agonists are listed in the Drug Summary Table at the end of this chapter.

Thromboxane Antagonists

Both TxA_2 receptor antagonists and thromboxane synthase inhibitors could theoretically represent powerful and selective agents capable of inhibiting platelet activity and protecting against thrombosis and vascular disease. Thromboxane antagonists could serve as "super" platelet inhibitors in the management of patients with cardiovascular disease. TxA_2 receptor antagonists, unlike aspirin, would also be expected to block the vasoconstrictive action of the isoprostanes. Compounds such as **dazoxiben** and **pirmagrel** inhibit thromboxane synthase, and **ridogrel** is a TxA_2 receptor antagonist. These thromboxane antagonists have not yet found clinical utility, however, because the clinical benefit of these drugs is not significantly greater than that of aspirin, which is far less expensive.

Leukotriene Inhibition

Lipoxygenase Inhibition

Inhibition of 5-lipoxygenase has the potential to represent a major therapeutic modality in diseases involving leukotriene-mediated pathophysiology, including asthma, inflammatory bowel disease, and rheumatoid arthritis. Lipoxygenase inhibition is an attractive therapeutic approach in these diseases because leukotrienes are potent, locally acting mediators.

Several strategies are possible for the design of lipoxygenase inhibitors, based on the structure, function, and mechanism of the lipoxygenase enzymes. Suicide inhibitors of lipoxygenase (e.g., derivatives of arachidonic acid with triple bonds instead of double bonds), which become covalently bound to the active site and render it inactive, have been developed but are not available for clinical use. Radical scavengers such as catechols, butylated hydroxytoluene (BHT), and α-tocopherol trap the radical intermediates in the lipoxygenase reaction and thereby prevent the functioning of the enzyme, but these nonspecific compounds cannot be used clinically for lipoxygenase inhibition.

Drugs that impair or alter the ability of lipoxygenase to utilize its nonheme iron would be expected to inhibit the activity of the enzyme. The only lipoxygenase inhibitor in clinical use is **zileuton** (Fig. 42-10A), a benzothiophene derivative of N-hydroxyurea that inhibits 5-LOX by chelating its nonheme iron. In asthma, zileuton induces bronchodilation, improves symptoms, and generates long-lasting improvement in pulmonary function tests. Zileuton is effective in the treatment of asthma induced by cold, drugs, and allergens. However, because of its low bioavailability, low potency, and significant adverse effects such as liver toxicity,

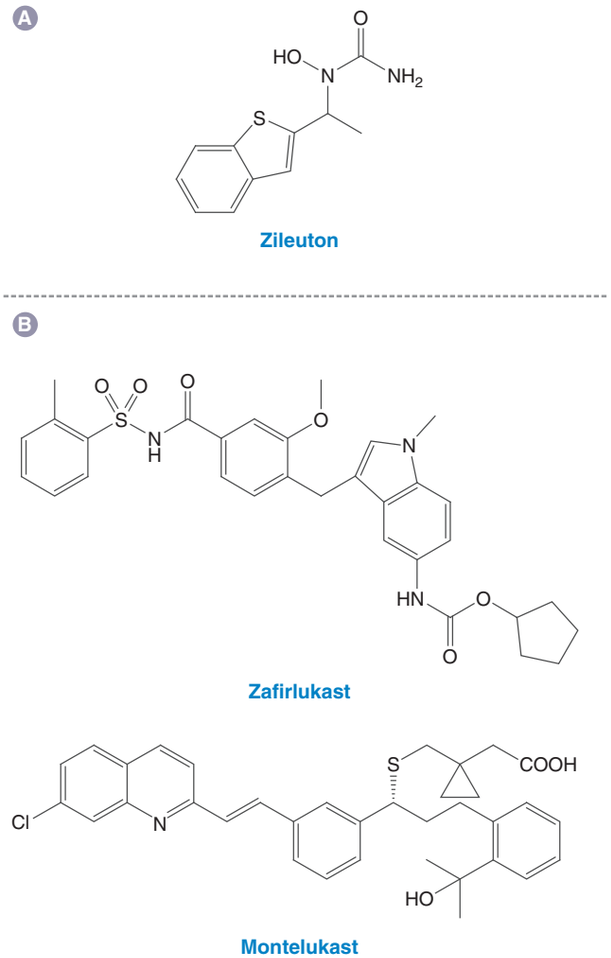


FIGURE 42-10. Leukotriene pathway inhibitors. A. Zileuton is a 5-lipoxygenase inhibitor that blocks the biosynthesis of leukotrienes from arachidonic acid. **B.** Zafirlukast and montelukast are $CysLT_1$ receptor antagonists. All three drugs are approved for the prophylaxis and chronic treatment of asthma in both adults and children. None of these drugs is effective in the treatment of acute asthma attacks.

zileuton is not as widely used as the other antileukotriene asthma drugs (see below).

5-Lipoxygenase Activating Protein (FLAP) Inhibition

Interfering with the role of FLAP could represent an alternative approach to the selective inhibition of 5-LOX activity and leukotriene function. Recall that 5-LOX is activated after the enzyme translocates to the nuclear membrane and docks with FLAP and that FLAP binds arachidonic acid released by phospholipase A_2 and shuttles it to the 5-LOX active site. FLAP inhibitors have been developed that both prevent and reverse LOX binding to FLAP and block the arachidonic acid binding site, but no FLAP inhibitors are currently available for clinical use.

Leukotriene Synthesis Inhibitors

Other than zileuton, no specific inhibitors of the enzymes involved in leukotriene synthesis are available for clinical use. Specific LTA_4 hydrolase inhibitors, which block LTB_4 biosynthesis, are currently in development. **Adenosine**, acting via its receptors on neutrophils, inhibits LTB_4 biosynthesis by regulating arachidonic acid release and, possibly, by interfering with the influx of calcium. Furthermore, adenosine is thought to have a role in limiting cell and tissue injury during inflammation. High cell turnover at inflammatory sites generates high local concentrations of adenosine, which may decrease LTB_4 biosynthesis and reduce leukocyte recruitment and activation. Selective adenosine receptor agonists could be considered for development as pharmacologic agents in the control of inflammation.

Leukotriene Receptor Antagonists

Leukotriene receptor antagonism represents a receptor-based mechanism for inhibiting leukotriene-mediated bronchoconstriction and other effects (Fig. 42-4). Cysteinyl leukotriene receptor (CysLT1) antagonists are effective against asthma induced by antigen, exercise, cold, or aspirin. These agents significantly improve bronchial tone, pulmonary function tests, and asthma symptoms. **Montelukast** and **zafirlukast** (Fig. 42-10B) are the currently available cysteinyl leukotriene receptor antagonists; the main clinical application for these antagonists is in the treatment of asthma.

More potent CysLT1 antagonists are in development, including pobilukast, tomelukast, and verlukast. Further research will likely elucidate cysteinyl leukotriene receptor subtypes and their respective tissue distributions, which could offer the possibility of tissue-targeted antagonism and the application of these tissue-selective antagonists to other conditions such as rheumatoid arthritis, inflammatory bowel disease, and various allergic disorders.

Lipoxins, Aspirin-Triggered Lipoxins, Resolvins/Protectins/Maresins, and Lipoxin-Stable Analogues

Lipoxins, ATLs, and the omega-3-derived resolvins, protectins, and maresins all offer the potential to antagonize the inflammatory actions of leukotrienes and other inflammatory mediators and to promote resolution of inflammation. Stable, oral and parenteral analogues of these compounds could represent a new approach to treatment, since they are agonists of endogenous anti-inflammation and pro-resolution pathways rather than direct enzyme inhibitors or receptor antagonists. Because lipoxins are endogenous regulators, they would be expected to have selective actions with few adverse effects. Stable analogues of lipoxins and ATLs are currently being developed, and second-generation lipoxin-stable analogues have shown efficacy in enhancing the resolution of recurring bouts of acute inflammation in skin inflammation and gastrointestinal inflammation models. This new approach to the treatment of inflammation remains to be established in human trials.

CONCLUSION AND FUTURE DIRECTIONS

Eicosanoids are critical mediators of homeostasis and of many pathophysiologic processes, especially those involving host defense and inflammation. Arachidonic acid is the important substrate and is converted into prostaglandins, thromboxanes, prostacyclin, leukotrienes, lipoxins, isoprostanes, and epoxyeicosatetraenoic acids (EETs). Prostaglandins have diverse roles in vascular tone regulation, gastrointestinal regulation, uterine physiology, analgesia, and inflammation.

Prostacyclin and thromboxane coordinately control vascular tone, platelet activation, and thrombogenesis. Leukotrienes (LTC_4, LTD_4) are the chief mediators of bronchoconstriction and airway hyperactivity; LTB_4 is a major activator of leukocyte chemotaxis and infiltration. Lipoxins antagonize the effects of leukotrienes, reduce the extent of inflammation, and activate resolution pathways.

Pharmacologic interventions at many critical points in these pathways are useful in limiting inflammatory sequelae. Glucocorticoids inhibit several steps in eicosanoid generation, including the rate-determining step involving phospholipase A_2. However, chronic glucocorticoid use is associated with many serious adverse effects, including osteoporosis, muscle wasting, and abnormal carbohydrate metabolism. Cyclooxygenase inhibitors block the first step of prostanoid synthesis and prevent the generation of prostanoid mediators of inflammation. Lipoxygenase inhibitors, FLAP inhibitors, leukotriene synthesis inhibitors, and leukotriene receptor antagonists prevent leukotriene signaling, thereby limiting inflammation and its deleterious effects. Future drug development efforts will allow selective targeting of eicosanoid pathways involved in many clinical conditions.

Systems biology has revealed mechanisms underlying inflammatory disease and has created a new discipline of resolution pharmacology. Essential omega-3 fatty acids, in particular eicosapentaenoic acid and docosahexaenoic acid, are precursors to pro-resolving and anti-inflammatory SPMs (specialized pro-resolving mediators) that serve a physiologic role leading to programmed resolution of inflammation (Fig. 42-6). These new bioactive mediators are many times more potent than their respective omega-3 precursors and, hence, may mediate the essential and beneficial effects of omega-3 fatty acids. In the near future, resolvins and protectins may be developed as new therapeutic agents to promote resolution of inflammation.

Suggested Reading

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Gilroy DW, Perretti M. Aspirin and steroids: new mechanistic findings and avenues for drug discovery. *Curr Opin Pharmacol* 2005;5:1–7. (*Reviews the anti-inflammatory actions of aspirin-triggered lipoxins and the discovery of annexin and related compounds in the actions of glucocorticoids.*)

Patrono C, Baigent C. Low-dose aspirin, coxibs, and other NSAIDS: a clinical mosaic emerges. *Mol Interv* 2009;9:31–39. (*Reviews the data regarding cardiovascular risks of NSAIDs and coxibs.*)

Psaty BM, Furberg CD. COX-2 inhibitors—lessons in drug safety. *N Engl J Med* 2005;352:1133–1135. (*Reviews issues surrounding withdrawal of COX-2 selective inhibitors.*)

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Serhan CN, Chiang N, Van Dyke TE. Resolving inflammation: dual anti-inflammatory and pro-resolution lipid mediators. *Nat Rev Immunol* 2008;8:249–261. (*Reviews advances in the role of eicosanoid pathways and novel lipid mediators in resolution programs of inflammation.*)

Vane JR, Bakhle YS, Botting RM. Cyclooxygenases 1 and 2. *Ann Rev Pharmacol Toxicol* 1998;38:97–120. (*Historic overview of prostaglandin research, including discussion of the pharmacologic manipulation of these pathways.*)

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DRUG SUMMARY TABLE: CHAPTER 42 Pharmacology of Eicosanoids

DRUG	CLINICAL APPLICATIONS	*SERIOUS* AND COMMON ADVERSE EFFECTS	CONTRAINDICATIONS	THERAPEUTIC CONSIDERATIONS
NONSTEROIDAL ANTI-INFLAMMATORY DRUGS (NSAIDs) *Mechanism—Inhibit cyclooxygenase-1 (COX-1) and cyclooxygenase-2 (COX-2), decreasing the biosynthesis of downstream eicosanoids and thereby limiting the inflammatory response*				
Aspirin	Mild to moderate pain Headache, myalgia, arthralgia Prophylaxis of stroke and myocardial infarction (antiplatelet effect)	*Gastrointestinal ulcer, bleeding, Reye's syndrome, asthma exacerbation, bronchospasm, angioedema* Gastrointestinal disturbance, tinnitus	Aspirin hypersensitivity Aspirin-triggered asthma Children and teenagers with chickenpox or flu symptoms, due to risk of Reye's syndrome	The oldest of the NSAIDs Widely used to treat mild to moderate pain, headache, myalgia, and arthralgia In contrast to other NSAIDs, aspirin acts in an irreversible manner by acetylating the active site serine residue in both COX-1 and COX-2 Aspirin increases plasma concentration of acetazolamide, which leads to CNS toxicity Ibuprofen may inhibit the antiplatelet effect of aspirin Limited reports suggest that salicylates may enhance methotrexate toxicity Aspirin increases the risk of bleeding in anticoagulated patients
Propionic acids: **Ibuprofen** **Naproxen** **Ketoprofen** **Flurbiprofen** ***Acetic acids:*** **Indomethacin** **Sulindac** **Etodolac** **Diclofenac** **Ketorolac** ***Oxicams:*** **Piroxicam** ***Fenamates:*** **Mefenamate** **Meclofenamate** ***Ketones:*** **Nabumetone**	Mild to moderate pain Fever Osteoarthritis, rheumatoid arthritis Dysmenorrhea Gout Patent ductus arteriosus closure (indomethacin)	*Gastrointestinal hemorrhage, ulceration, perforation; nephrotoxicity; Stevens–Johnson syndrome; pseudoporphyria (naproxen)* Gastrointestinal disturbance, tinnitus	Gastrointestinal or intracranial bleeding Coagulation defects Asthma, urticaria, or allergic-type reactions after taking NSAIDS, due to risk of severe, even fatal, anaphylactic reactions Significant renal insufficiency	Naproxen has a longer half-life, is 20 times more potent, and causes fewer gastrointestinal adverse effects than aspirin Ketorolac is used for analgesia in postsurgical patients; it is used for no more than 3–5 days Piroxicam has a long half-life; once-daily dosing Nabumetone has the greatest selectivity of these agents for COX-2 Fenamates have limited use; compared to aspirin, fenamates have less anti-inflammatory activity and higher toxicity

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DRUG SUMMARY TABLE: CHAPTER 42 Pharmacology of Eicosanoids *continued*

DRUG	CLINICAL APPLICATIONS	*SERIOUS* AND COMMON ADVERSE EFFECTS	CONTRAINDICATIONS	THERAPEUTIC CONSIDERATIONS
ACETAMINOPHEN *Mechanism—Weak inhibitor of peripheral cyclooxygenases; predominant effect may be inhibition of cyclooxygenase-3 (COX-3) in the CNS*				
Acetaminophen	Fever Mild to moderate pain	*Hepatotoxicity, nephrotoxicity (rare)* Rash, hypothermia	Hypersensitivity to acetaminophen	Although acetaminophen has analgesic and antipyretic effects similar to aspirin, the anti-inflammatory effect of acetaminophen is insignificant because of its weak inhibition of peripheral cyclooxygenases Generally safe for use in patients undergoing surgery and dental procedures May inhibit COX-3 isoform in CNS Acetaminophen overdose is a leading cause of hepatic failure Antidote for acetaminophen overdose is N-acetylcysteine
COX-2 SELECTIVE INHIBITORS *Mechanism—Selective inhibition of COX-2*				
Celecoxib	Osteoarthritis, rheumatoid arthritis in adults, and ankylosing spondylitis Primary dysmenorrhea Acute pain in adults Familial adenomatous polyposis	*Myocardial infarction, ischemic stroke, heart failure, gastrointestinal bleeding, ulceration, perforation; renal papillary necrosis; exacerbation of asthma* Gastrointestinal disturbance, peripheral edema	Hypersensitivity to sulfonamides Hypersensitivity to celecoxib Asthma, urticaria, or allergic-type reactions after taking NSAIDS, due to risk of severe, even fatal, anaphylactic reactions Pain associated with coronary artery bypass graft surgery	Decreases efficacy of ACE inhibitors Incidence of gastropathy and nephropathy may be less than that associated with NSAIDs, but may still be significant Valdecoxib and rofecoxib recently withdrawn from U.S. market due to possible increase in cardiovascular mortality
GLUCOCORTICOIDS *Mechanism—Inhibit COX-2 action and prostaglandin biosynthesis by inducing lipocortins, activating endogenous anti-inflammatory pathways, and other mechanisms*				
Prednisone **Prednisolone** **Methylprednisolone** **Dexamethasone**	See Drug Summary Table: Chapter 28 Pharmacology of the Adrenal Cortex			
CYTOKINE ANTAGONISTS *Mechanism—Etanercept, infliximab, adalimumab, and golimumab inhibit TNF-α; anakinra inhibits IL-1*				
Etanezrcept **Infliximab** **Adalimumab** **Golimumab**	See Drug Summary Table: Chapter 45 Pharmacology of Immunosuppression			
Anakinra	See Drug Summary Table: Chapter 45 Pharmacology of Immunosuppression			

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PROSTANOID MIMETICS *Mechanism—Prostanoid receptor agonists; see specific drug*				
Alprostadil	Maintenance of patent ductus arteriosus Erectile dysfunction	*Heart failure, cardiac arrhythmia and conduction defects, disseminated intravascular coagulation (DIC), disorders of bone development, seizure, priapism, apnea in newborn* Hypotension, penile fibrosis, penile discomfort	Sickle cell anemia or trait Leukemia, myeloma Neonatal respiratory distress syndrome Anatomical deformation of the penis, penile implant, Peyronie's disease	PGE_1 analogue with vasodilator properties Used primarily for maintaining patent ductus arteriosus in tetralogy of Fallot, Eisenmenger pulmonary hypertension, and aortic valve atresia
Misoprostol	Cytoprotective and antisecretory effects against gastric ulcers in long-term NSAID therapy Abortifacient with mifepristone	*Rare anemia, rare cardiac arrhythmia* Gastrointestinal disturbance	Pregnancy	PGE_1 analogue with vasodilator properties Also used in peptic ulcer disease (see Chapter 46) Cytoprotective effects likely mediated by increasing gastric mucus and bicarbonate production; antisecretory effects mediated via inhibition of basal and nocturnal gastric acid secretion by parietal cells
Carboprost	Abortion in second trimester Postpartum hemorrhage	*Dystonia, pulmonary edema* Gastrointestinal disturbance with prevalent diarrhea, headache, paresthesia, fever, breast tenderness	Acute pelvic inflammatory disease Cardiac, pulmonary, renal, or hepatic disease	$PGF_{2\alpha}$ analogue that stimulates uterine contraction for abortifacient activity; luteolytic activity controls fertility
Latanoprost **Bimatoprost** **Travoprost**	Ocular hypertension Open-angle glaucoma	*Macular retinal edema* Blurred vision, hyperpigmentation of eyelid, iris pigmentation	Hypersensitivity to latanoprost, bimatoprost, or travoprost	PGF_2a analogues with vasodilator properties; ocular hypotensive agents
Epoprostenol	Pulmonary hypertension	*Supraventricular tachycardia, hemorrhage, thrombocytopenia* Hypotension, rash, gastrointestinal disturbance, musculoskeletal pain, paresthesia, anxiety, influenza-like illness	Heart failure with severe left ventricular dysfunction Chronic use in patients developing pulmonary edema	Prostacyclin analogue that stimulates vasodilation of pulmonary and systemic arterial vasculature; also inhibits platelet aggregation
THROMBOXANE ANTAGONISTS *Mechanism—Inhibit thromboxane synthase or antagonize thromboxane receptor; investigational agents*				
Dazoxiben **Pirmagrel** **Ridogrel**	Dazoxiben and pirmagrel inhibit thromboxane synthase; ridogrel is a thromboxane A_2 receptor antagonist Advantages of these agents over aspirin are unproven Little effect on platelet aggregation			

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DRUG SUMMARY TABLE: CHAPTER 42 Pharmacology of Eicosanoids *continued*

DRUG	CLINICAL APPLICATIONS	*SERIOUS* AND COMMON ADVERSE EFFECTS	CONTRAINDICATIONS	THERAPEUTIC CONSIDERATIONS
LIPOXYGENASE INHIBITOR *Mechanism—Inhibits 5-lipoxygenase, which catalyzes the formation of leukotrienes from arachidonic acid*				
Zileuton	Asthma	*Increased liver enzymes* Urticaria, abdominal discomfort, dizziness, insomnia	Active liver disease Elevated liver enzymes	Avoid concurrent use of dihydroergotamine, ergoloid mesylates, ergonovine, and methylergonovine due to an increased risk of ergotism (nausea, vomiting, vasospastic ischemia)
LEUKOTRIENE RECEPTOR ANTAGONISTS *Mechanism—Selective antagonists of the cysteinyl leukotriene (CysLT) type-1 receptor*				
Montelukast **Zafirlukast**	Chronic asthma Perennial allergic rhinitis (montelukast) Seasonal allergic rhinitis (montelukast)	*Allergic granulomatosis angiitis, hepatitis* Gastrointestinal distress, hallucinations, agitation	Hypersensitivity to montelukast or zafirlukast	Montelukast and zafirlukast are not indicated for acute asthma attacks and are generally not appropriate as monotherapy for asthma Both drugs are excreted in breast milk

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QUERY:
AQ1: Sentence beginning “15-HETE is converted by 5-lipoxygenase to the chemical intermediate...” correct as meant?

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